

Parotid and Temporal Regions, Infratemporal fossa and TMJ

1. Parotid Region

- is the posterolateral part of the facial region, bounded by the:
 - Zygomatic arch **superiorly**.
 - External ear and anterior border of the SCM **posteriorly**.
 - Ramus of the mandible **medially**.
 - Anterior border of the masseter muscle **anteriorly**.
 - Angle and inferior border of the mandible **inferiorly**

Boundaries of the Parotid Region

The Parotid region is the posterolateral part of the face

Boundaries:

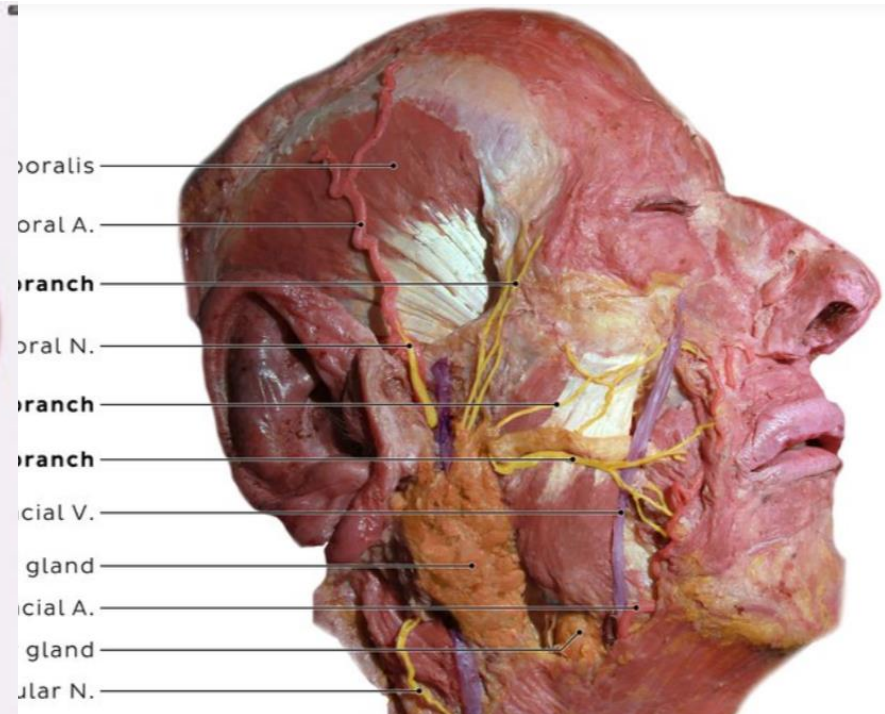
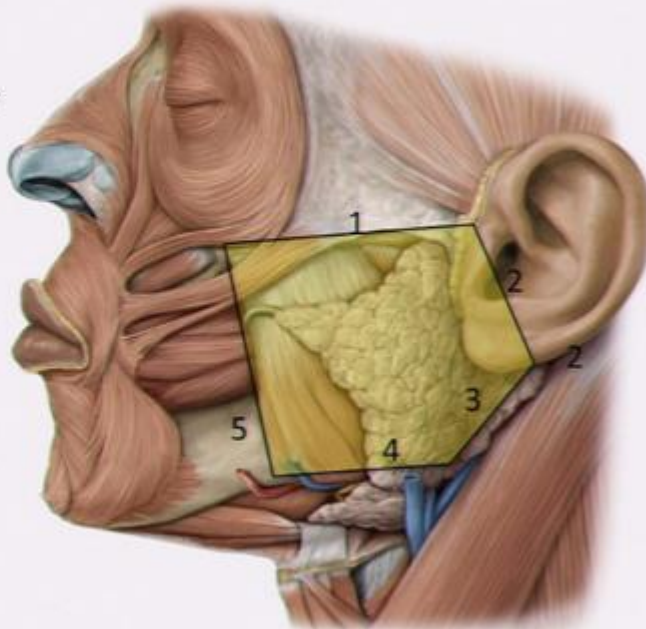
Zygomatic arch

External ear and anterior
Sternocleidomastoid muscle

Ramus of the mandible

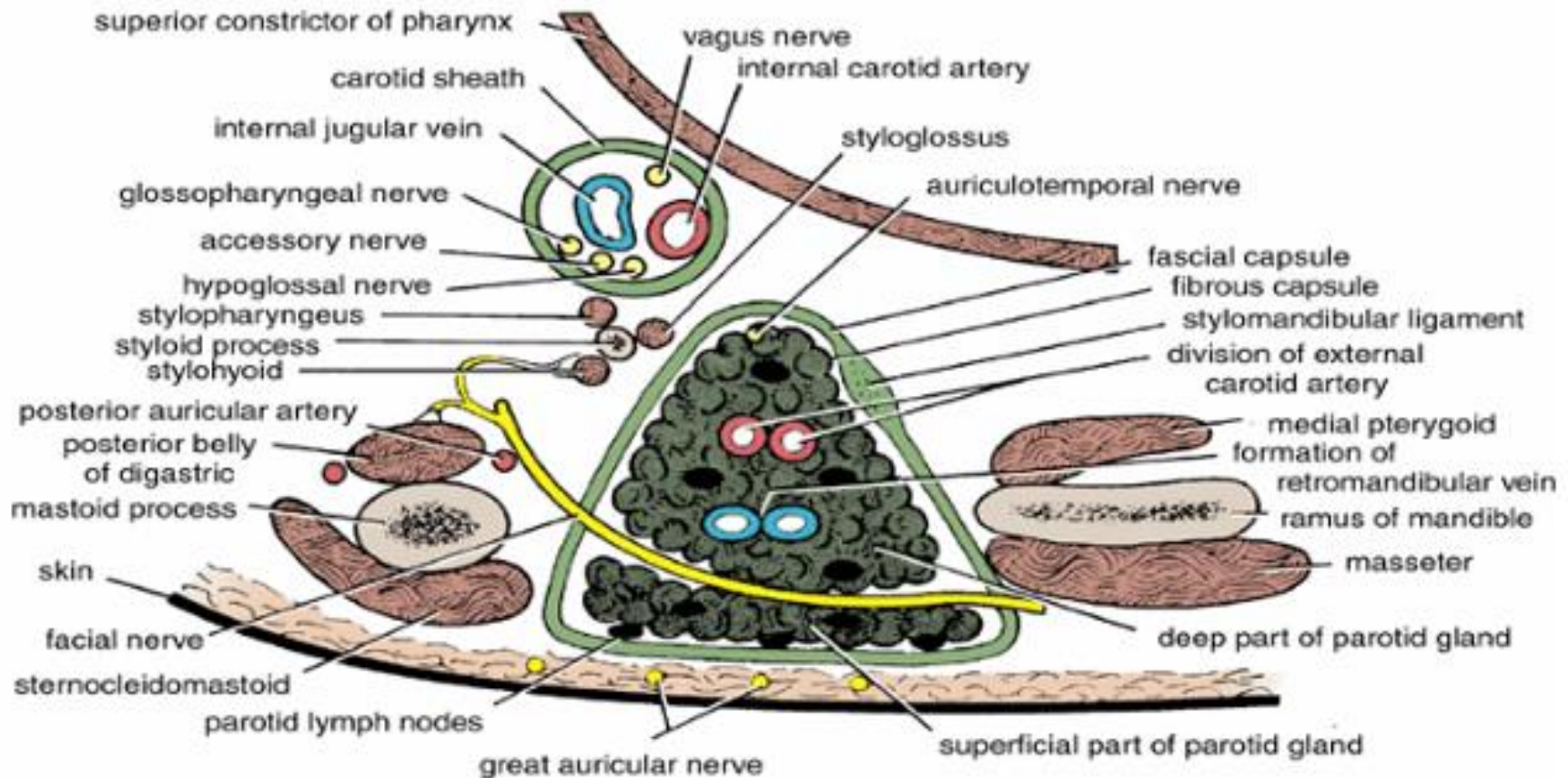
Angle and inferior border of
the mandible

Anterior border of the
masseter



• Contents:

- the parotid gland and duct
- the parotid plexus of the facial nerve (CN VII)
- the retromandibular vein
- the external carotid artery, and the masseter muscle

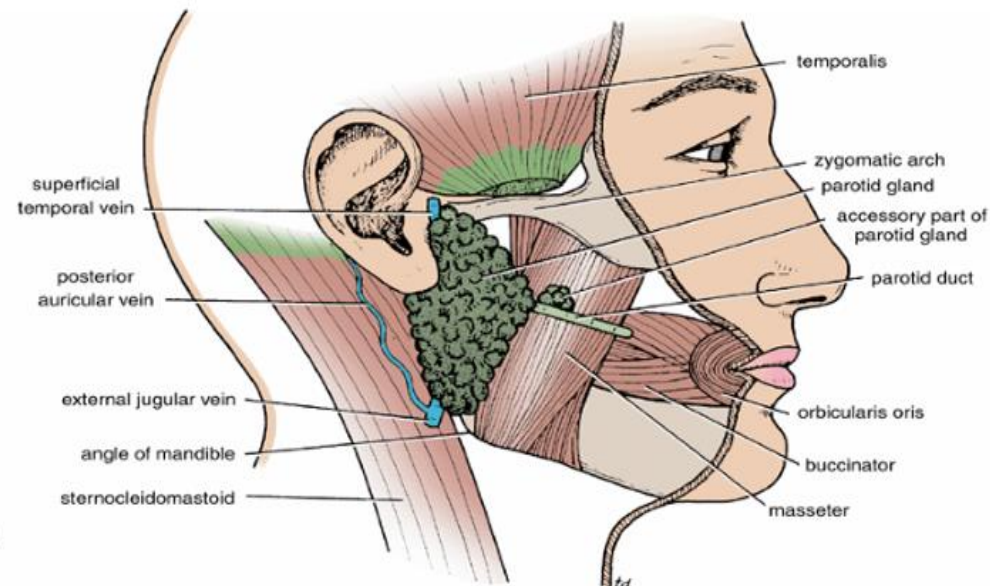


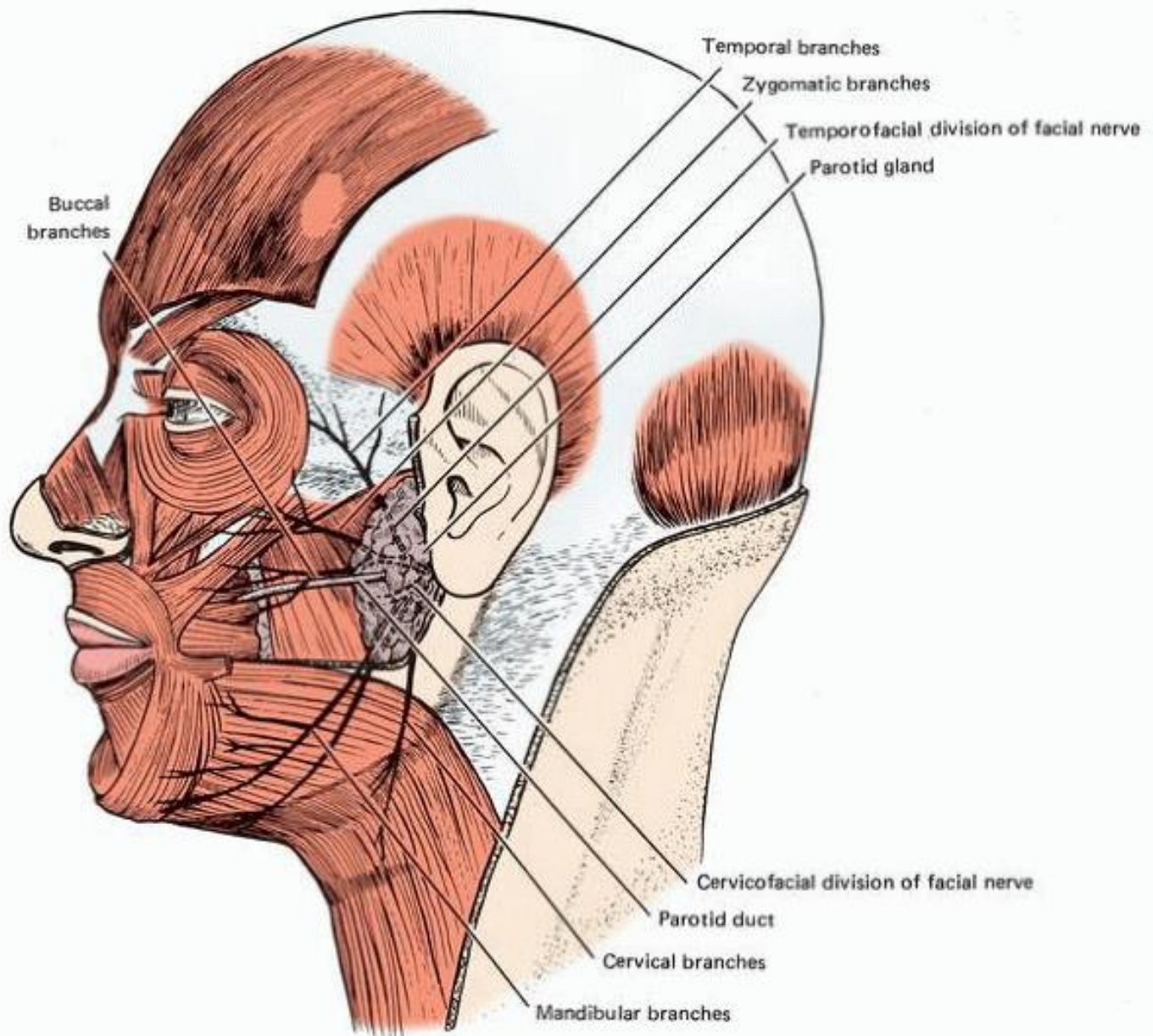
PAROTID GLAND

- the largest of three paired salivary glands.
- enclosed within a tough, fascial capsule (the **parotid sheath**), derived from the investing layer of deep cervical fascia
- has an irregular shape because the area occupied by the gland, **the parotid bed**:
 - is anteroinferior to the external acoustic meatus, where it is wedged between **the ramus of the mandible** and **the mastoid process**

The parotid bed:

- The space lying between:
 - the ramus of the mandible
 - external acoustic meatus
 - mastoid and styloid processes





- The medial extent of this bed :
 - is bordered by the posterior belly of the digastric muscle and the muscles attached to the styloid process
 - inferiorly, the bed is bounded by the superoanterior border of the sternocleidomastoid muscle.
 - this somewhat irregularly-shaped area houses the parotid gland, which is molded into this space, thus assisting in filling out the contour of the jaw/neck/ear junction

Parotid gland:

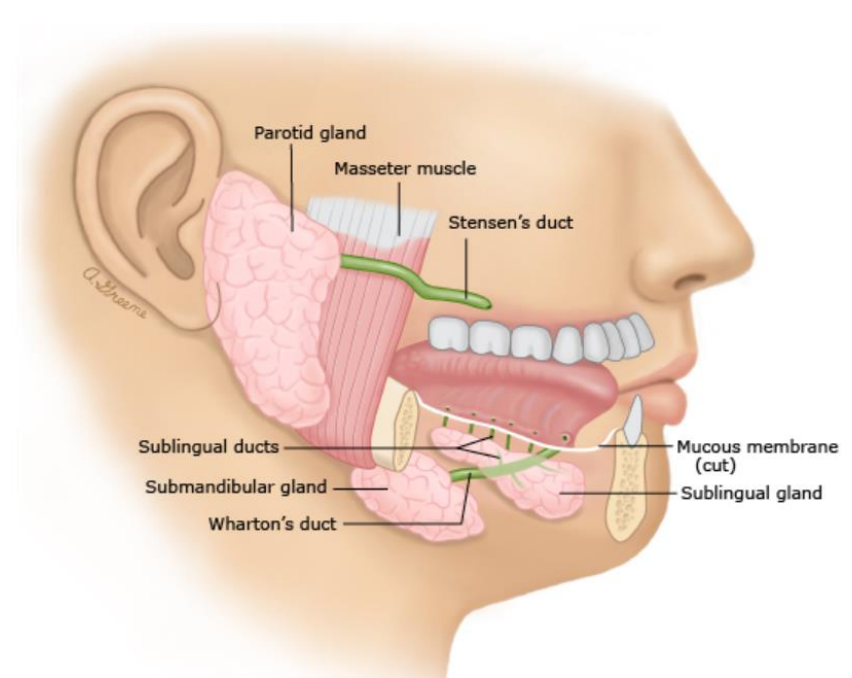
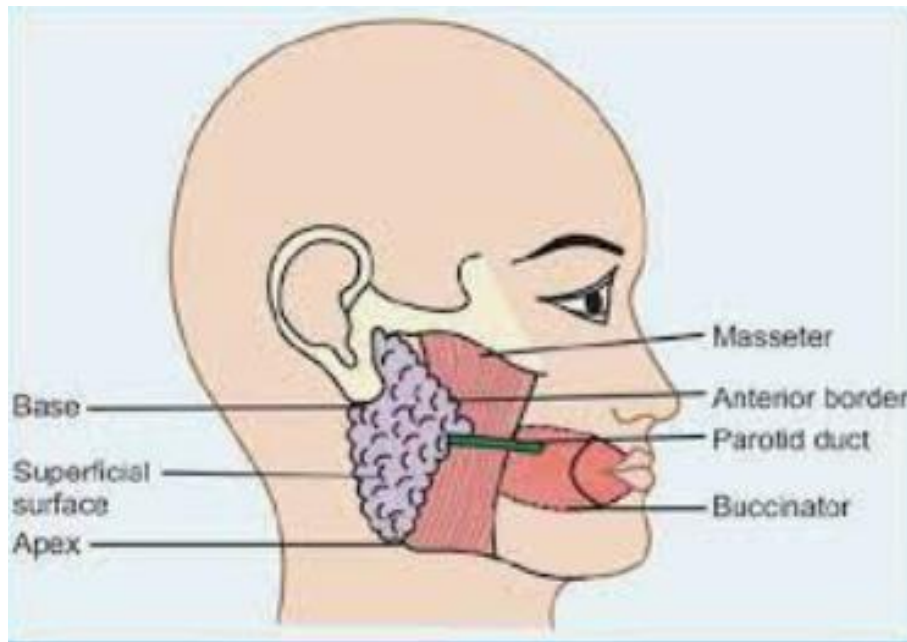
- Shape: **it is also irregular in shape**, because the gland is molded into an irregular space
- The superficial aspect of the gland extends:
 - **superiorly:**
 - over the masseter muscle to the zygomatic arch, where an accessory portion of the gland may be detached from the main substance.
 - **Inferiorly:**
 - it is mostly confined to the region between the mastoid process, the sternocleidomastoid muscle, and the angle of the mandible, where it extends over the posterior aspect of the masseter muscle

- **Medially:**

- the gland extends into the deeper portions of the parotid bed to the styloid process and **its attached musculature**.
- a wedge-shaped portion of the gland may intervene between the **medial and lateral pterygoid** muscles for a short distance.
- Often, glandular lobes extend into other spaces adjacent to the parotid bed
- One such lobe passes between the ramus of the mandible and the **medial pterygoid muscle** (above that muscle's insertion)
- whereas other lobes pass between the external auditory meatus and the temporomandibular joint (TMJ) and between the external carotid artery and the superior constrictor muscle of the pharynx

• Parotid duct (Stenson duct):

- is about 5 cm long and 5 mm in diameter.
- Exits the anterior aspect of the superficial portion of the gland
- passes anteriorly, superficial to the masseter muscle, to dive medially into the buccal fat pad
- it pierces the buccinator muscle on its way to the oral vestibule
- It delivers the parotid salivary secretions at the opening of the parotid papilla located **opposite the second maxillary molar**



- Contents of parotid gland:
 - Structures associated with the superficial aspect of the gland:
 - branches of the great auricular nerve
 - small lymph nodes that drain the superficial area
 - Structures associated with the deep aspect of the gland:
 - the external and internal carotid arteries
 - the internal jugular vein
 - both the vagus and glossopharyngeal nerves

- Several structures pass through the gland
 - The external carotid artery
 - enters the substance of the gland
 - it is there that several of its branches arise, such - the posterior auricular, maxillary, and superficial temporal arteries.
 - The retromandibular vein as well as the veins uniting to form it, also pass through the gland
 - The facial nerve (cranial nerve VII)
 - exits the stylomastoid foramen and enters the substance of the gland.
 - While in the gland, the facial nerve forms a plexus (**parotid plexes**) before exiting the gland to innervate the muscles of facial expression

- Structures passing through the parotid gland
 - The gland is separated into the superficial and deep lobes by the facial nerve (CN VII)
 - The nerve enters the posterior border of the parotid gland after leaving the stylomastoid foramen
 - Within the substance of the gland, CN VII bifurcates into the superior (or **temporofacial**) and inferior (**cervicofacial**) divisions
 - The part of the gland that is medial to CN VII is the deep lobe, while the part that is lateral to the nerve is the superficial lobe.

– The auriculotemporal nerve:

- a branch of the mandibular division of the trigeminal nerve (cranial nerve V)
- enters the substance of the gland from its deep aspect along the neck of the mandible
- emerges from the gland just inferior to the root of the zygomatic arch.
- While within the gland, it communicates with the facial nerve and distributes fibers to the gland

Remember the secretomotor pathway of the parotid gland with the mnemonic '**IT** has **L**esser **O**ptions **A**newhere", which stands for;

- **I**nferior salivatory nucleus
- **T**ympanic branch of 9th nerve
- **L**esser petrosal nerve
- **O**tic ganglion
- **A**uriculotemporal nerve

- The borders of the parotid gland are as follows:
 - Superiorly by the zygomatic arch
 - Posterosuperiorly by the external acoustic meatus and the ramus of the mandible
 - Posteroinferiorly by the mastoid process and distal part of the mastoid head of sternocleidomastoid muscle
 - Inferiorly the gland is limited by the sternocleidomastoid muscle
 - Anteriorly the masseter muscle forms another

• Has:

– The apex :

- posterior to the angle of the mandible

– Base:

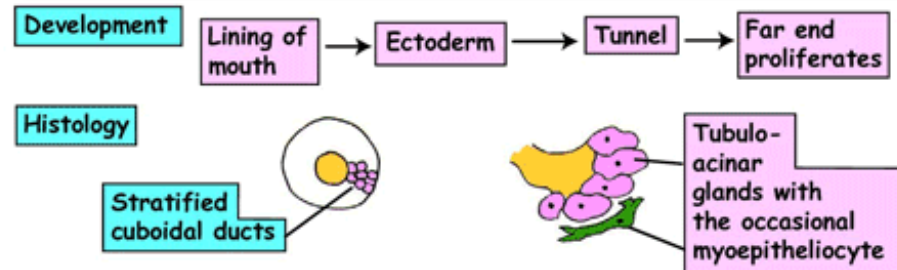
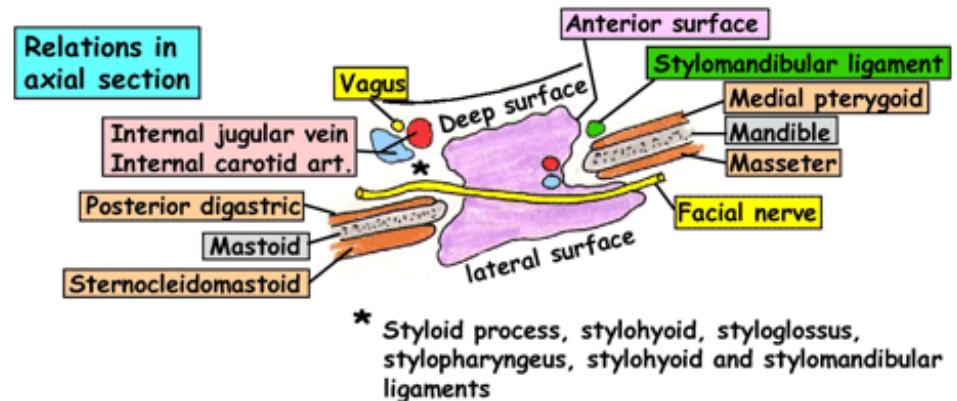
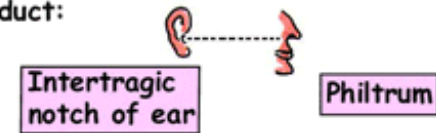
- is related to the zygomatic arch

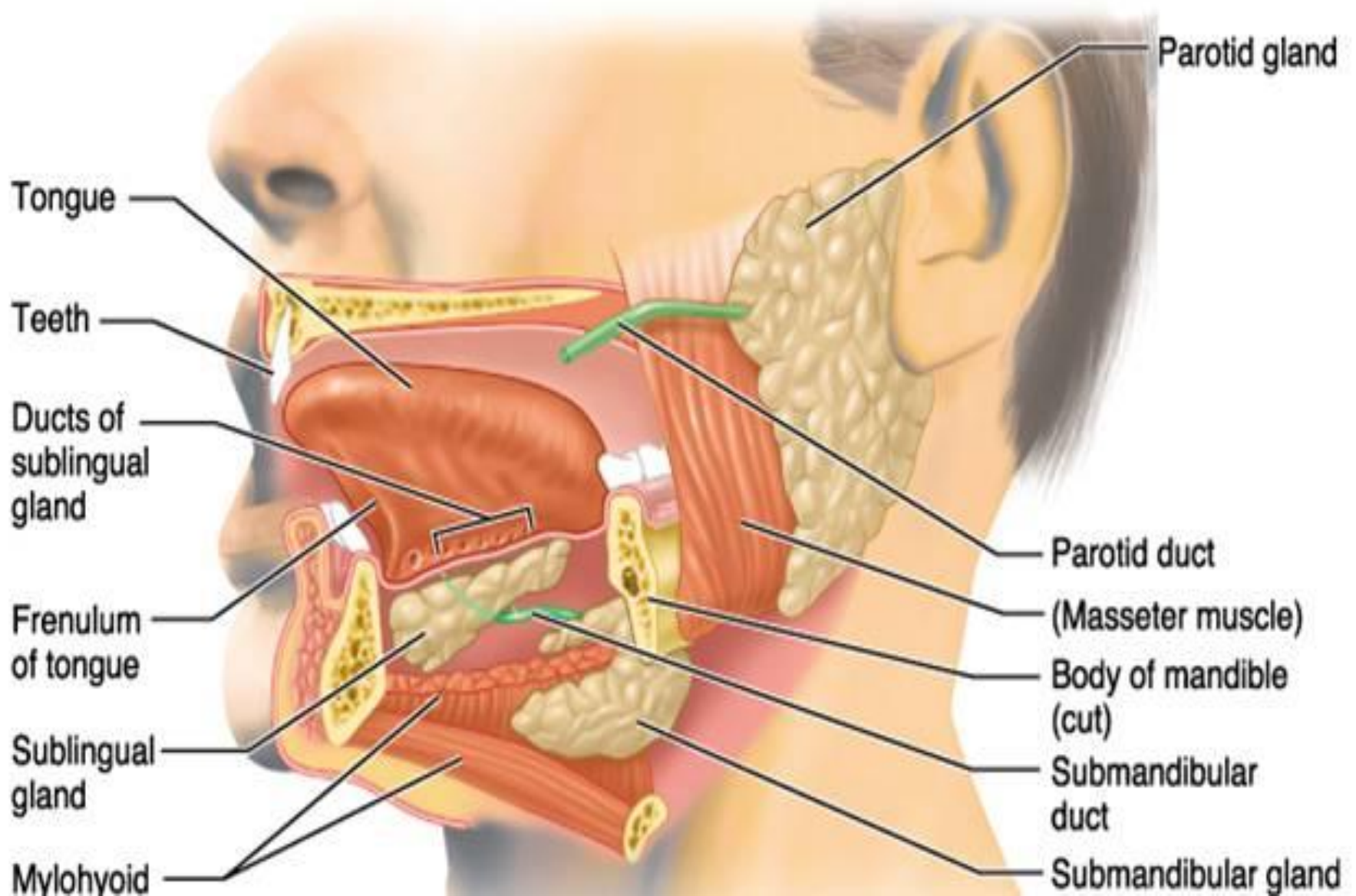
– Five surfaces:

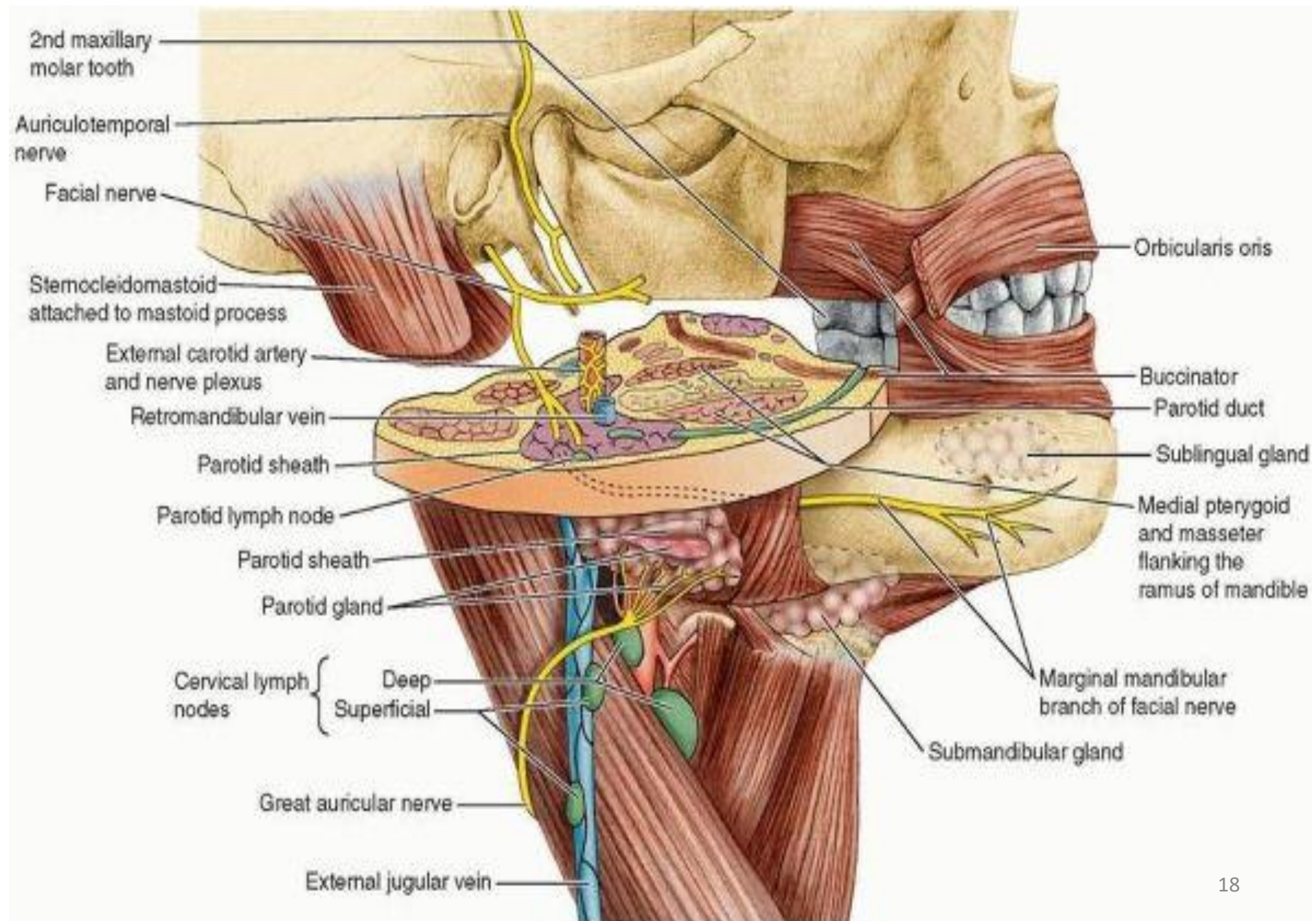
- Posterior
- Superior
- inferior,
- anterior surfaces
- And **lateral surface**

PAROTID GLAND 2

- Blood supply: Branches of external carotid
- Venous drainage: To retromandibular
- Lymph drainage: Pre-auricular to deep cervical
- Nerve supply: **Secretomotor** via inferior salivary nucleus to glossopharyngeal nerve to its tympanic branch to lesser petrosal nerve to otic ganglion to auriculotemporal nerve. **Sympathetics** via superior cervical ganglion and external carotid artery. **Sensation** for gland - auriculotemporal (Vc), for fascia - great auricular (C2)
- Duct: 5cm long, crosses masseter, pierces buccinator at 3rd molar and mucosa at 2nd molar. Stenson's duct
- Surface markings of duct:







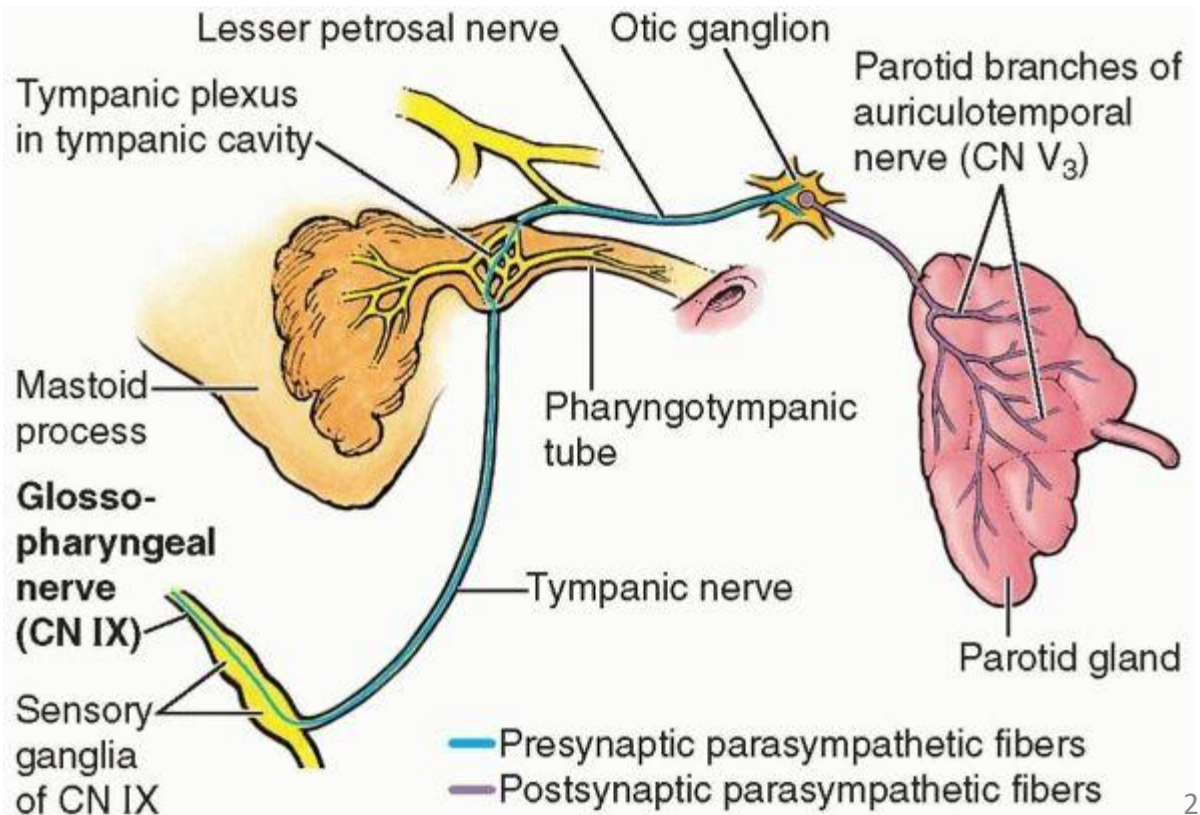
- **Clinical Considerations**
 - An inflamed **parotid papilla** (the orifice of the parotid duct in the oral cavity) may be an indication that the problem may be caused by the parotid gland rather than by a tooth
 - Diseases of the parotid gland cause pain to be referred to **the ear, the TMJ, and the external auditory meatus.**
 - This is due to the overlapping of sensory branches of various nerves serving the regions of the **parotid bed, ear, and TMJ**

Vascular supply

- The posterior auricular artery
- superficial temporal and transverse facial arteries
- Venous drainage is via the tributaries passing through the gland, and these vessels empty into the external jugular vein

Innervations

- the great auricular nerve
- the external carotid nerous plexus
- the glossopharyngeal nerve
- the auriculotemporal nerve



Temporal Region

- Includes:
 - the lateral area of the scalp and the deeper soft tissues overlying the temporal fossa of the cranium
 - superior to the zygomatic arch
- The temporal fossa:
 - occupied primarily by the upper portion of the temporalis muscle
 - is bounded:
 - Posteriorly and superiorly by the temporal lines.
 - Anteriorly by the frontal and zygomatic bones
 - Laterally by the zygomatic arch.
 - Inferiorly by the infratemporal crest

Articular tubercle

Greater wing of sphenoid

Zygomatic process of
temporal bone

Zygomatic bone

Infratemporal } Surface of
Anterior maxilla

Coronoid process

Mandibular notch

Squamous part of
temporal bone

Supramastoid crest

Postglenoid tubercle

External acoustic
meatus

Suprameatal
spine

Tympanomastoid
fissure

Tympanic part of
temporal bone

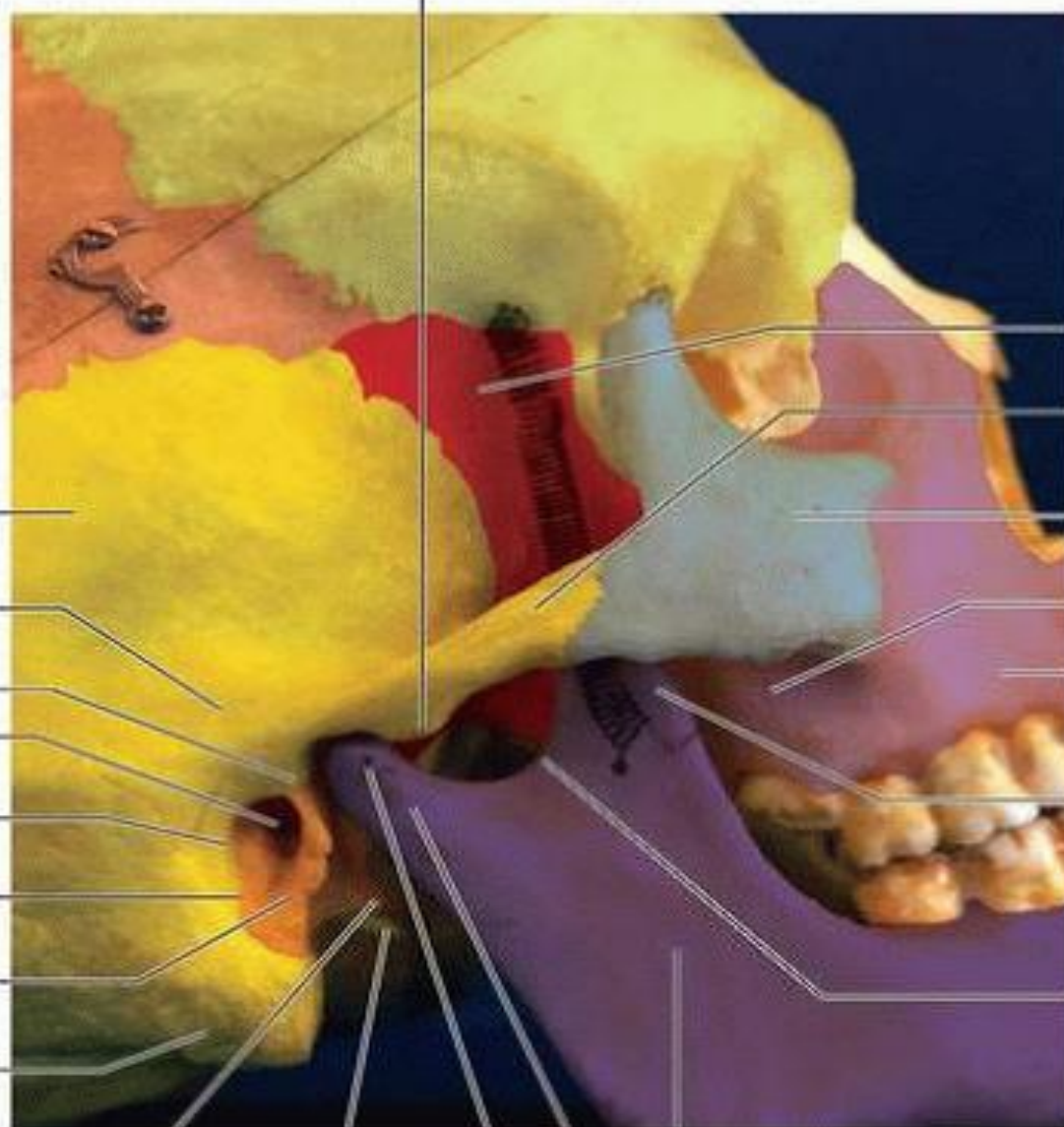
Mastoid proces

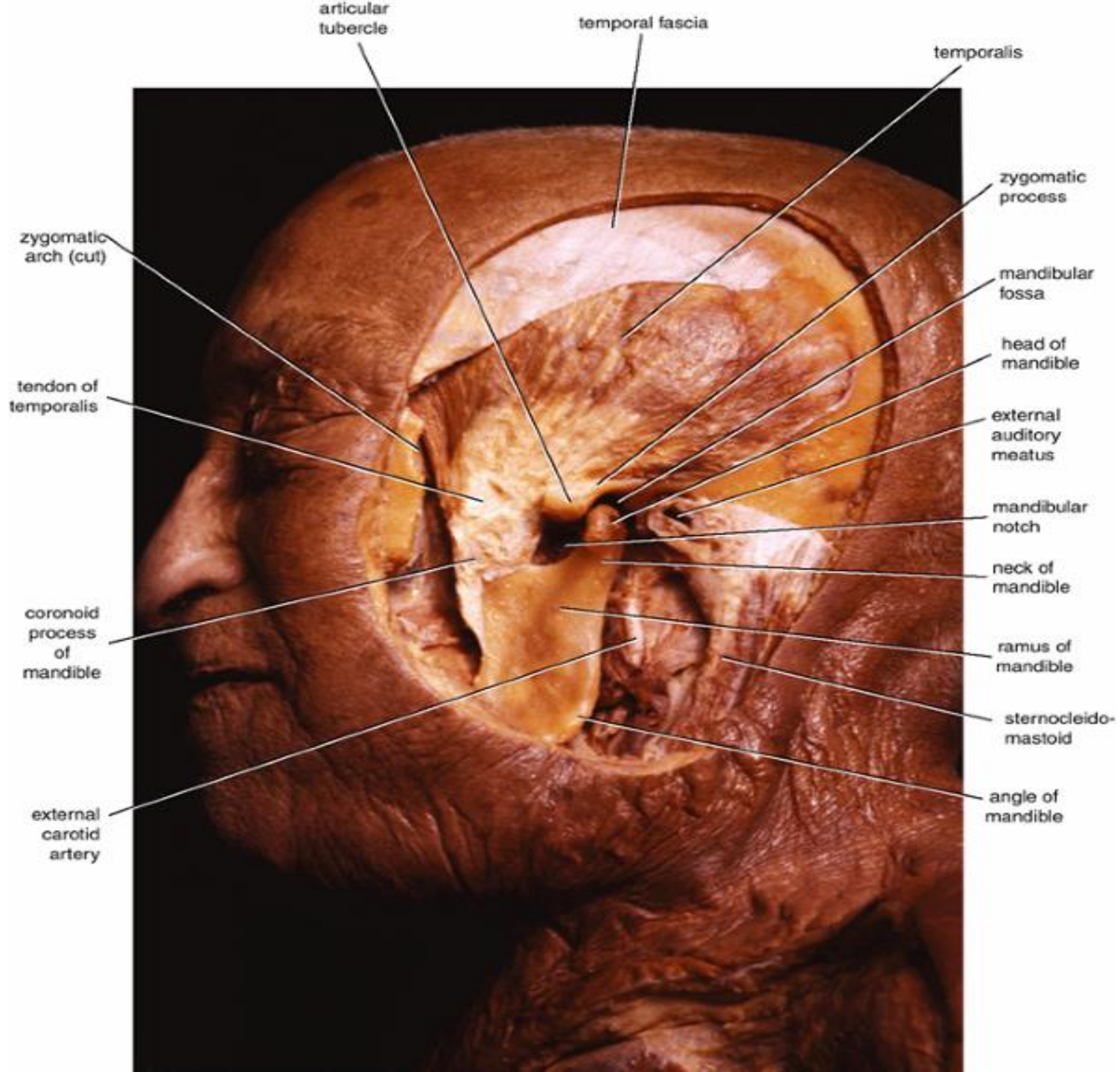
Lateral view

Vaginal
process

Styloid
process

Head Neck Ramus
Of mandible

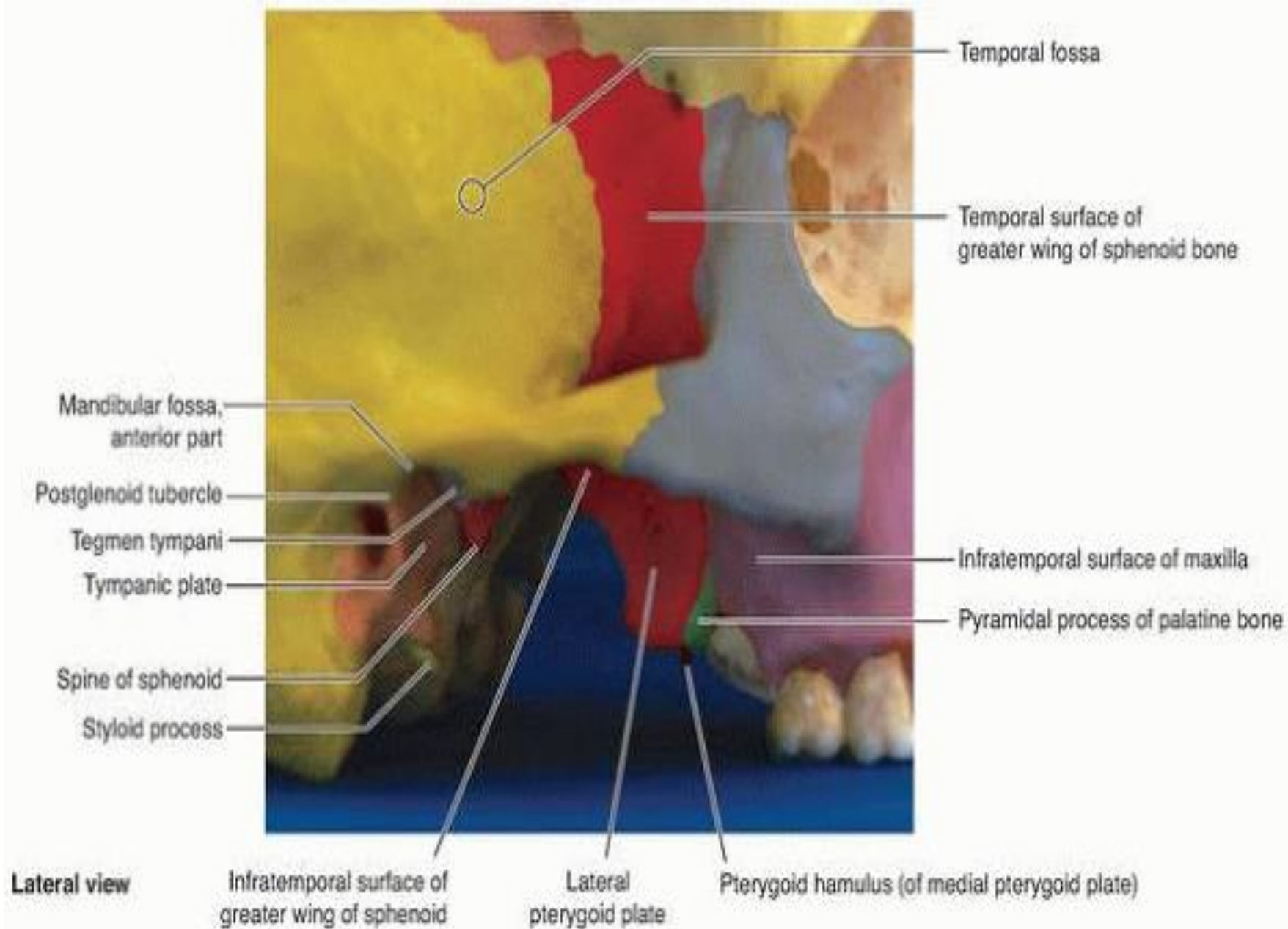


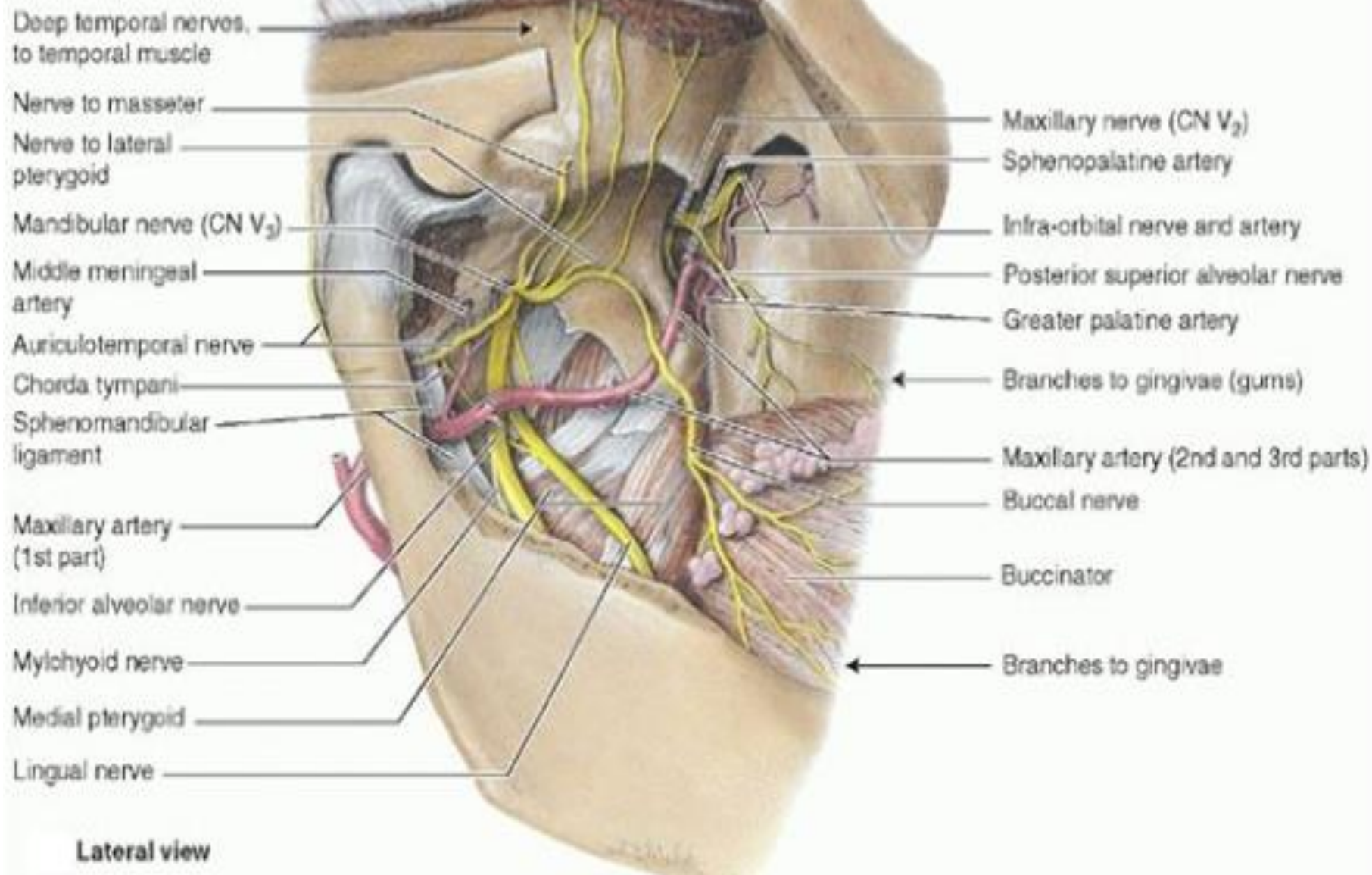


- The floor:
 - parts of the four bones that form the pterion
- Roof:
 - temporal fascia:
 - Extends superiorly to superior temporal line
 - Inferiorly, the fascia splits into two layers
 - attach to the lateral and medial surfaces of the zygomatic arch
 - Covers the temporalis muscle
 - When the powerful masseter muscle, which is attached to the inferior border of the arch, contracts and exerts a strong downward pull on the zygomatic arch, the temporal fascia provides resistance

Infratemporal Fossa

- an irregularly shaped space
 - deep and inferior to the zygomatic arch
 - deep to the ramus of the mandible
 - posterior to the maxilla
- communicates with the temporal fossa:
 - through the interval b/n (deep to) the zygomatic arch and (superficial to) the cranial bones.



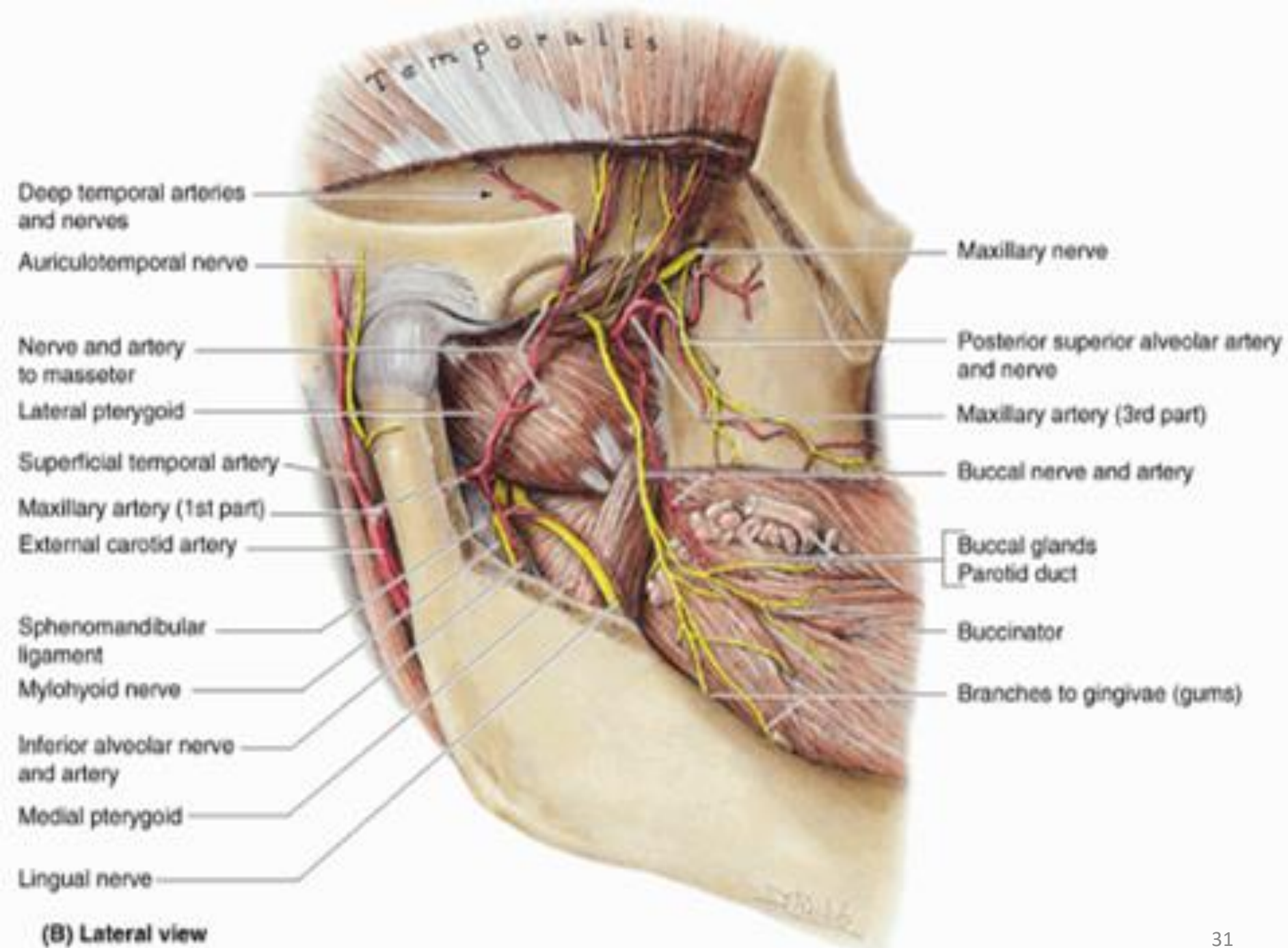


The boundaries:

- **Laterally:** the ramus of the mandible.
- **Medially:** the lateral pterygoid plate.
- **Anteriorly:** the posterior aspect of the maxilla.
- **Posteriorly:** the tympanic plate and the mastoid and styloid processes of the temporal bone.
- **Superiorly:** the inferior (infratemporal) surface of the greater wing of the sphenoid.
- **Inferiorly:** where the medial pterygoid muscle attaches to the mandible near its angle

contains:

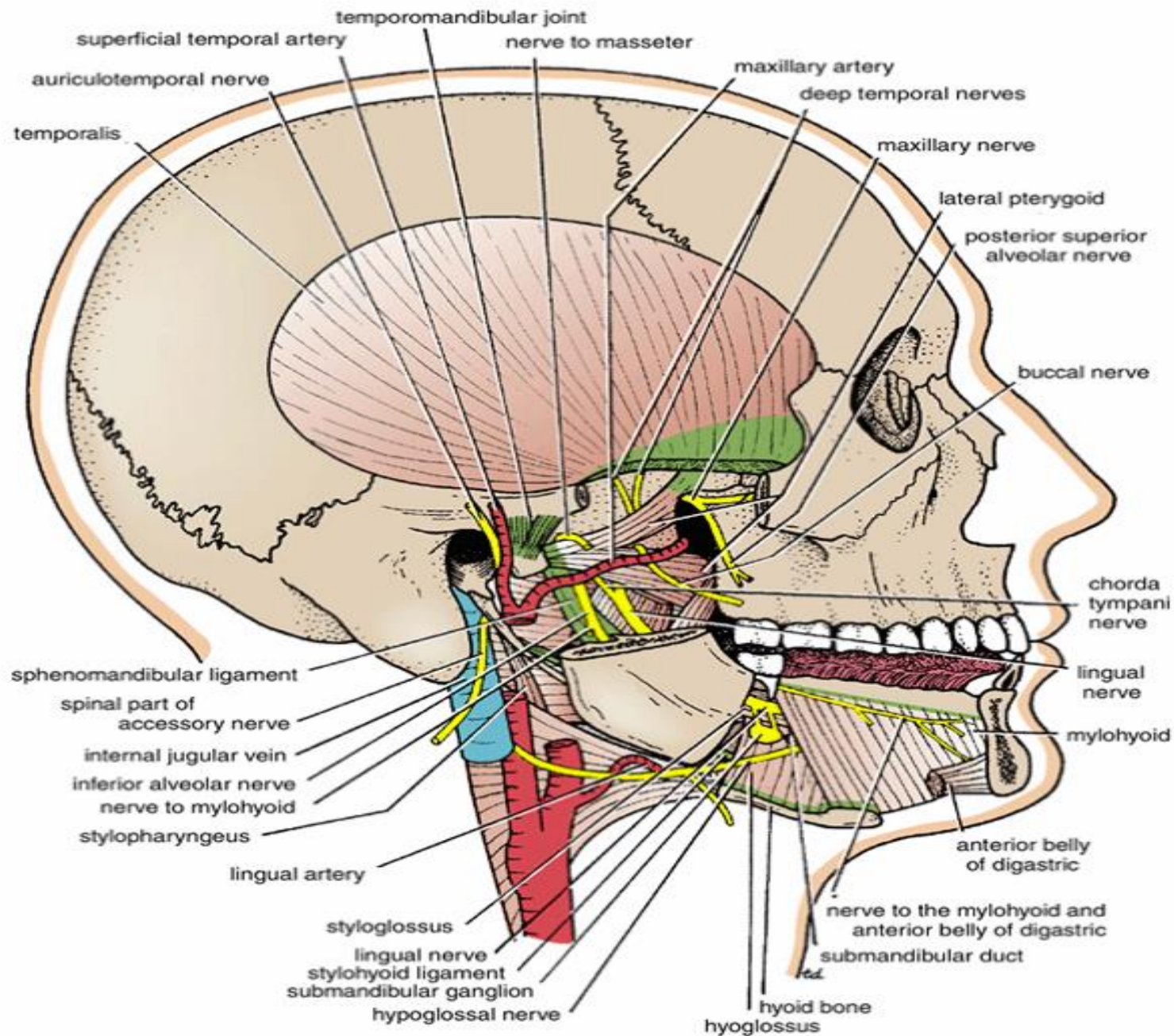
- Inferior part of the temporalis muscle.
- Lateral and medial pterygoid muscles.
- Maxillary artery.
- Pterygoid venous plexus.
- Mandibular, inferior alveolar, lingual, buccal, and chorda tympani nerves
- Otic ganglion.

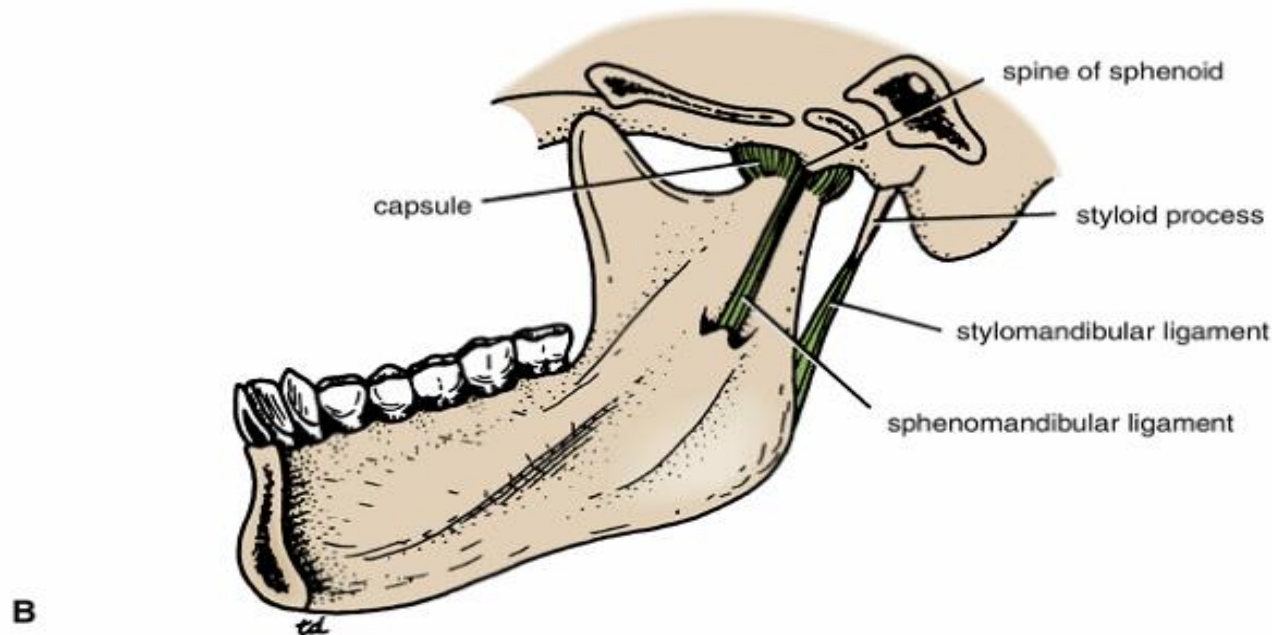
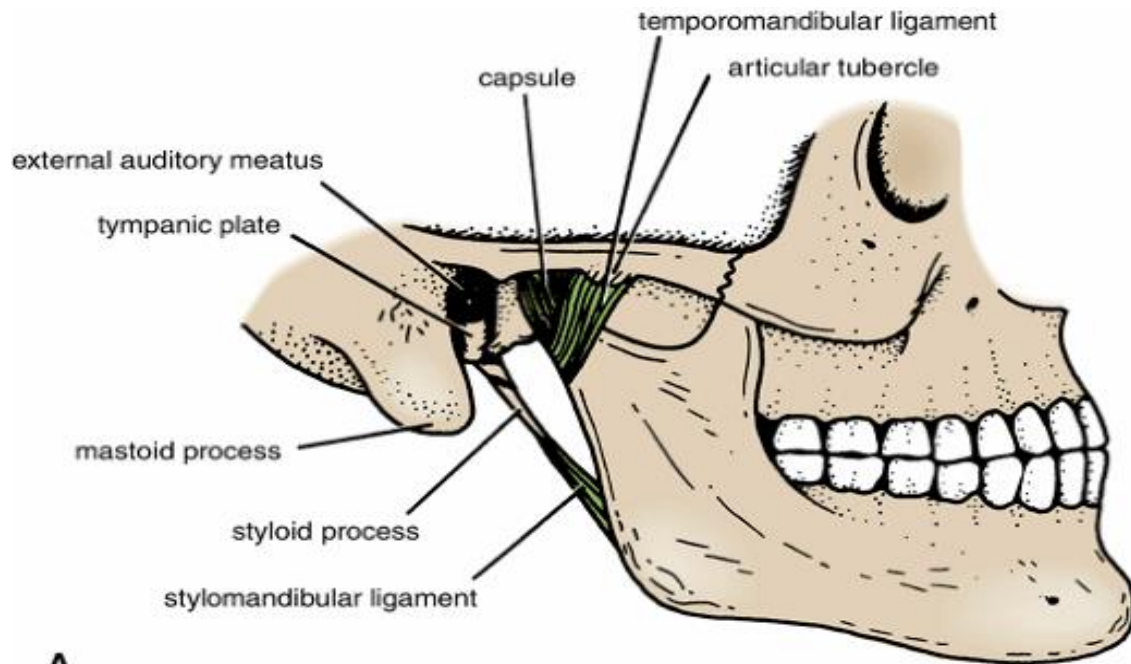


- **Temporomandibular Joint (TMJ):**
 - is the site of articulation between:
 - the mandible and the skull, specifically the area about the **articular eminence** of the temporal bone
 - this bilateral joint functions to open and close the jaws
 - to approximate the teeth of the opposing arches during mastication
 - consists of parts of **the mandible** and temporal bones
 - covered by dense, fibrous connective tissue and are surrounded by several ligaments.
 - Interposed between the two bones is a **fibrous articular disc**

Important relations of the TMJ

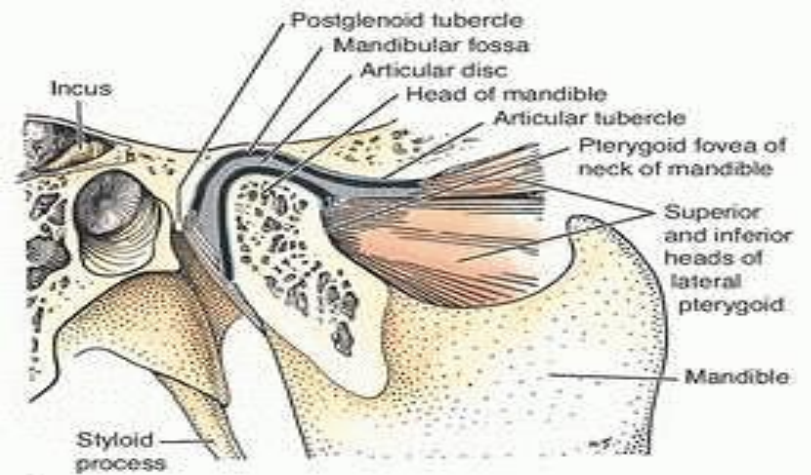
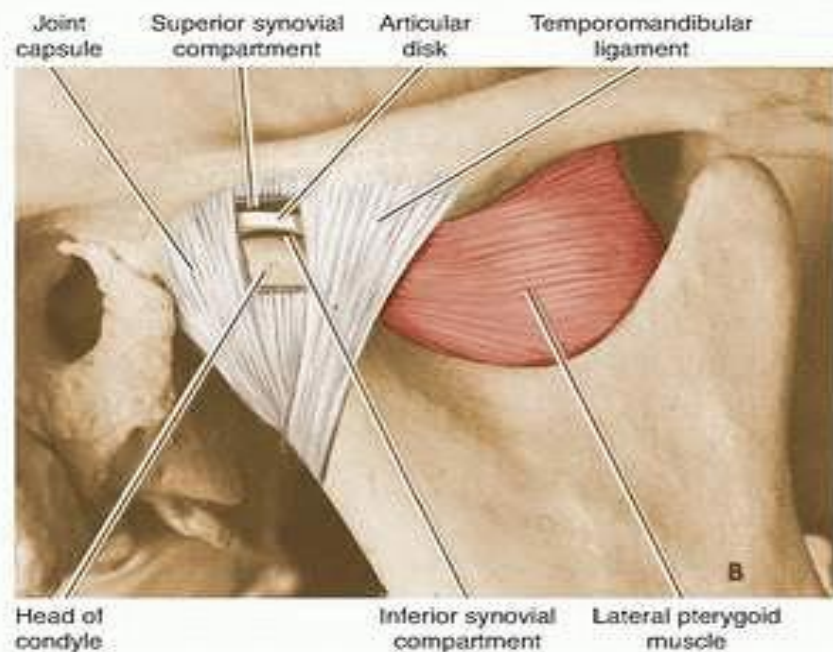
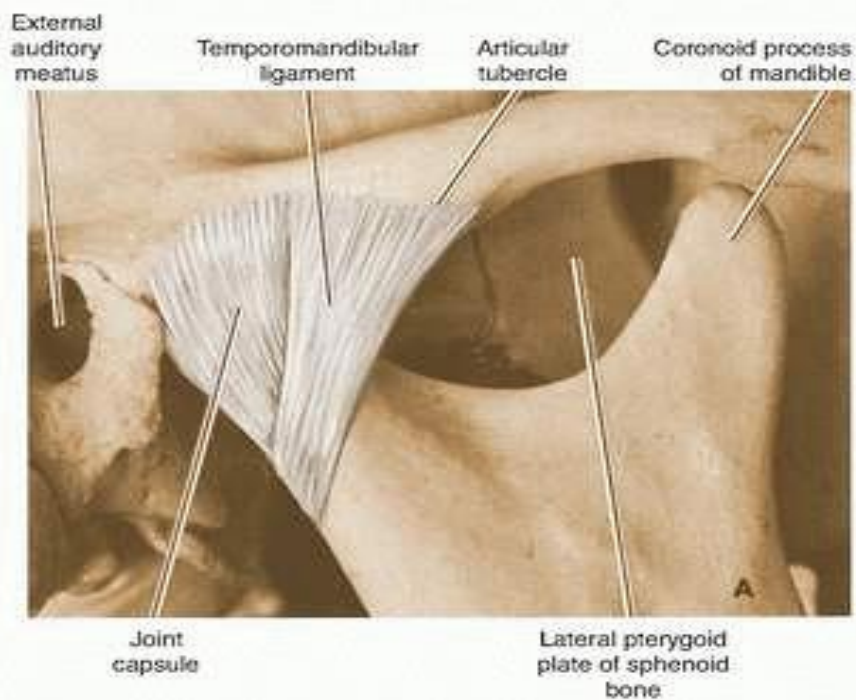
- Anteriorly:
 - The mandibular notch and the masseteric nerve and artery
- Posteriorly:
 - the tympanic plate of the external auditory meatus
 - the **glenoid process** of the parotid gland
- Laterally:
 - The parotid gland, fascia, and skin
- Medially:
 - The maxillary artery and vein and the auriculotemporal nerve



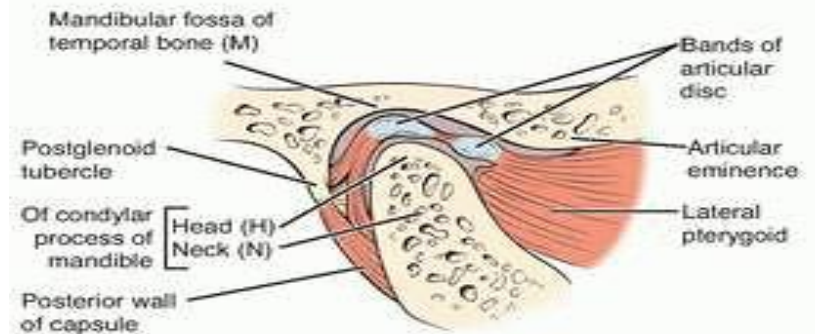


Joint Anatomy

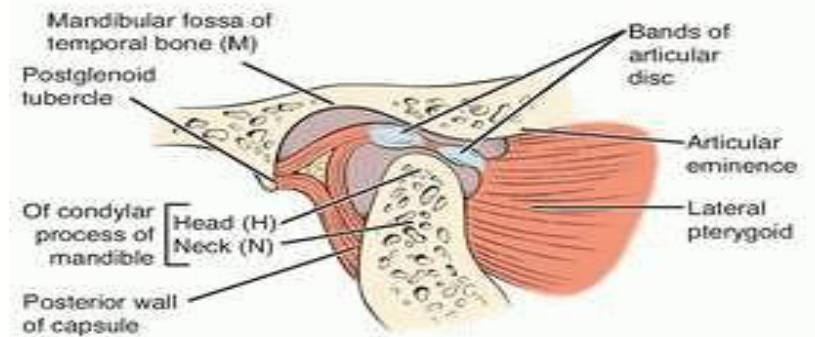
- Mandible
 - is the only freely moving bone of the skull.
 - possesses **two condylar processes** whose condyle articulates with a disc that is pressed against the **articular eminence** of the **temporal bone**
- The mandible possesses:
 - two articular surfaces, a condyle (head) located on the superior extremity of each of the bilateral condylar processes.
 - Each condyle articulates with a disc that is interposed between it and the temporal bone



A Sagittal section



B



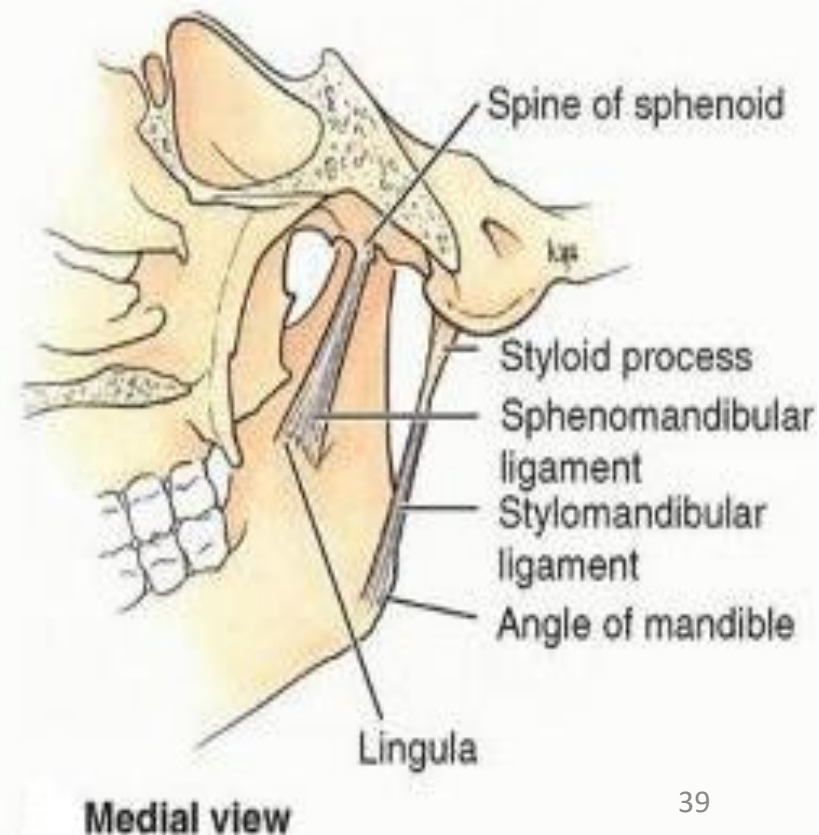
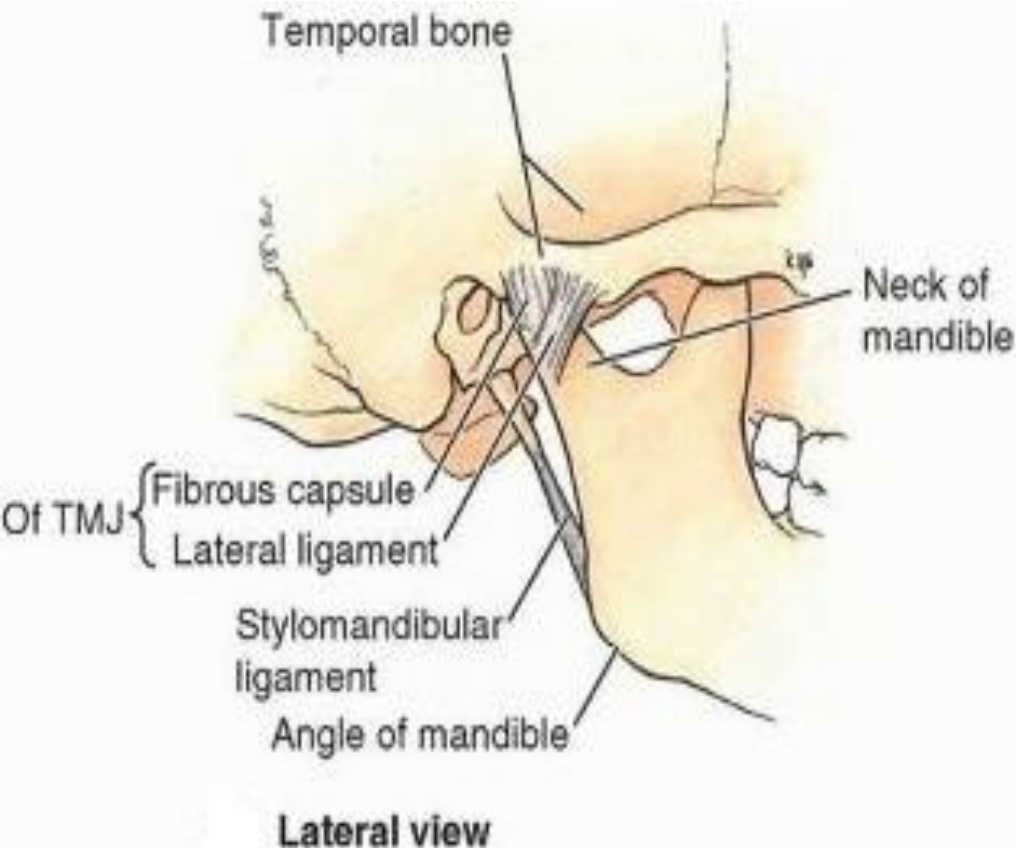
C

- Temporal Bone
 - The site of TMJ articulation on the temporal bone is on the **inferior surface of the zygomatic process**
 - The specific location is situated on the posterior slope of **the articular eminence**
- Articular Coverings
 - are composed of dense, collagenous connective tissue overlying a thin proliferative layer of cells associated with the underlying hyaline cartilage.
- Capsule
 - The TMJ capsule, composed of dense, irregular collagenous connective tissue, encloses the entire articulating region of the temporal bone, disc, and mandibular condyle, sealing the joint space

• Ligaments

- the lateral ligament or temporomandibular ligament
- sphenomandibular ligament
- The stylomandibular ligament

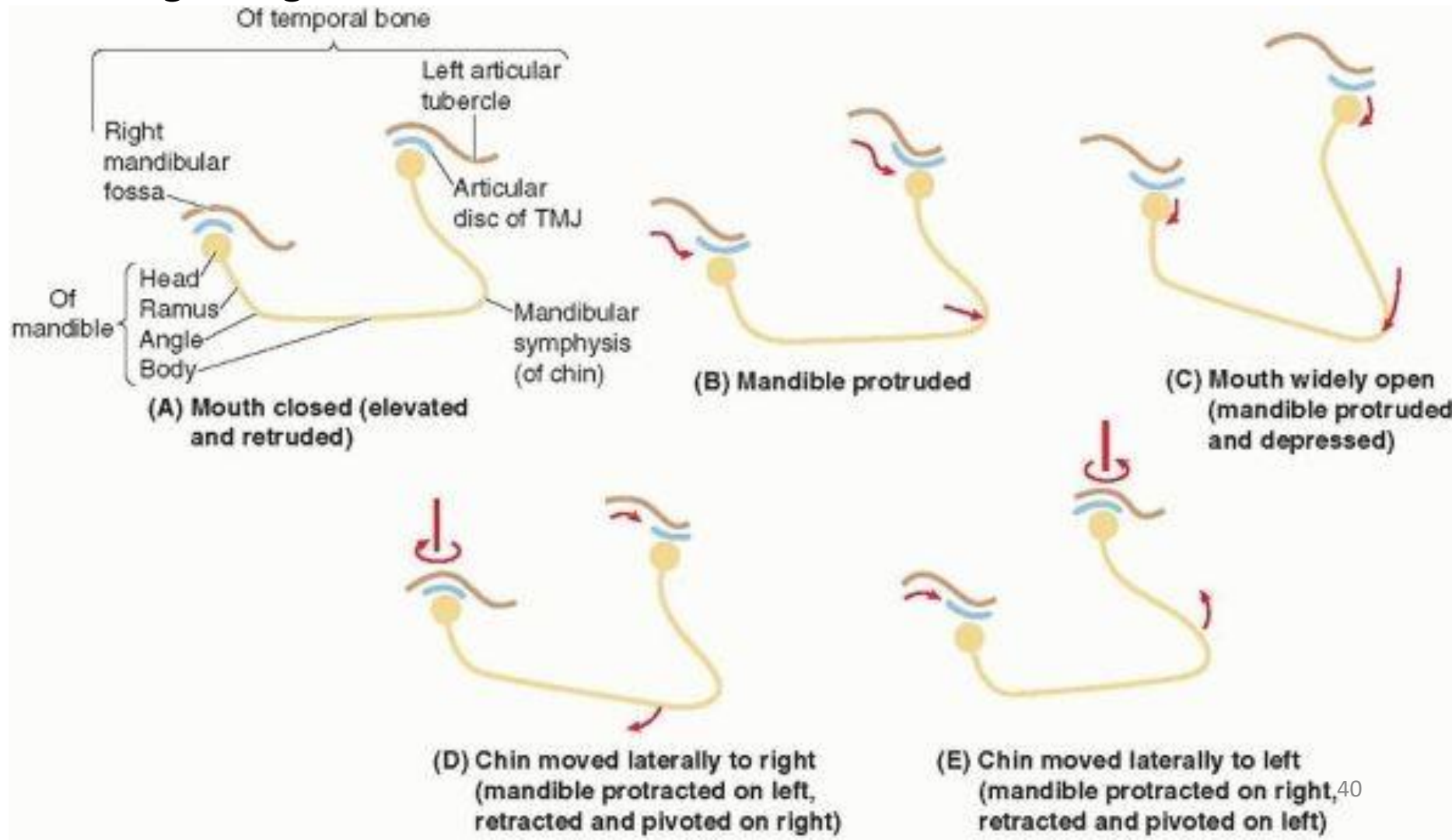
accessory ligament



• TYPES OF MOVEMENT

– of two types

- Hinge
- gliding

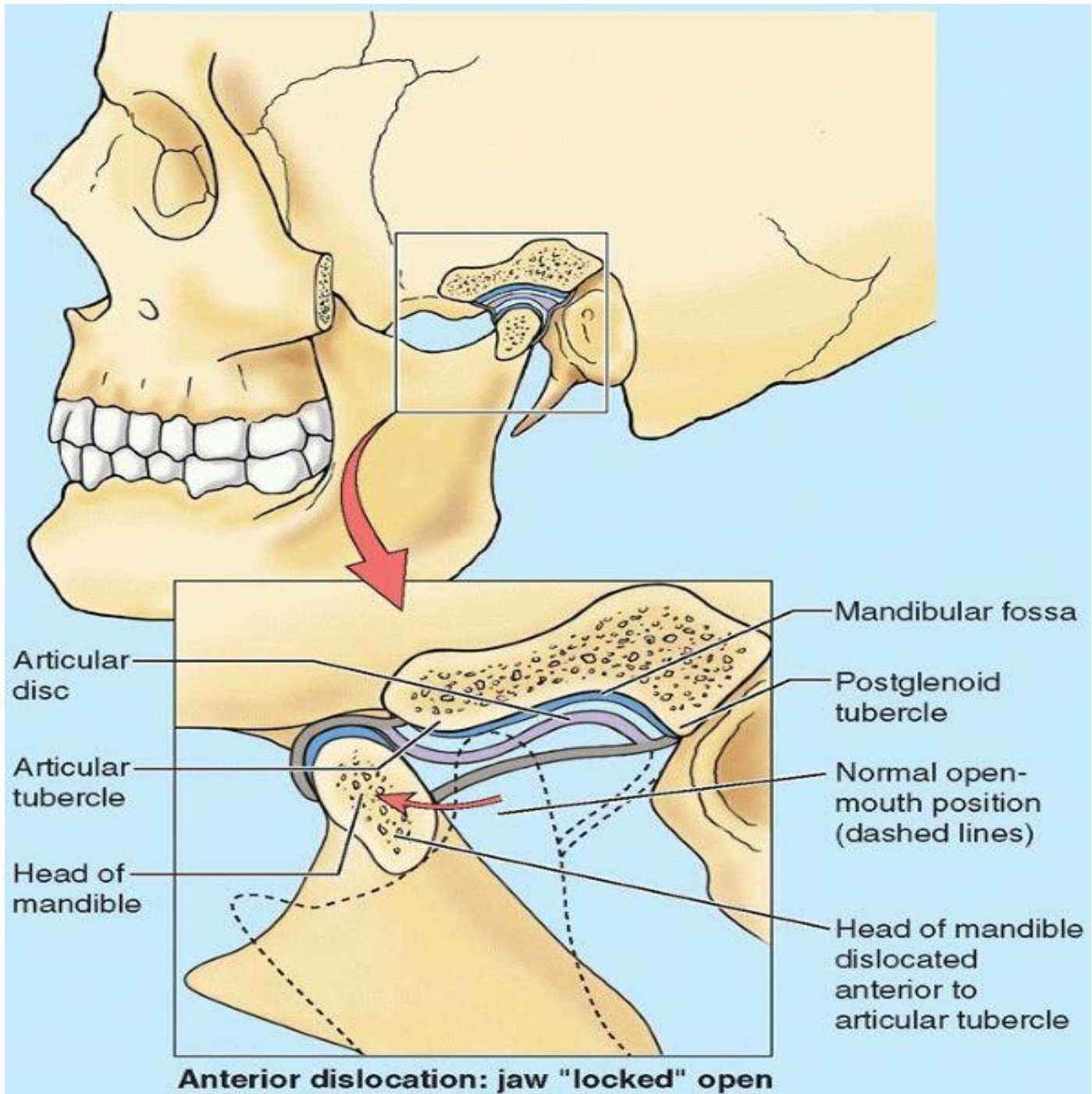


MOVEMENTS OF TEMPOROMANDIBULAR JOINT

<i>Movements of Mandible</i>	<i>Muscle(s)</i>
Elevation (close mouth)	Temporalis, masseter, and medial pterygoid
Depression (open mouth)	Lateral pterygoid and suprahyoid and infrahyoid muscles ^a
Protrusion (protrude chin)	Lateral pterygoid, masseter, and medial pterygoid ^b
Retrusion (retrude chin)	Temporalis (posterior oblique and near horizontal fibers) and masseter
Lateral movements (grinding and chewing)	Temporalis of same side, pterygoids of opposite side, and masseter

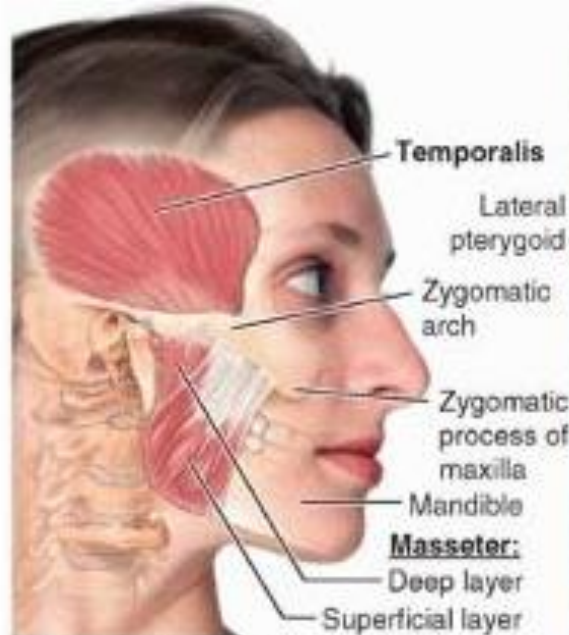
- **Dislocation of the TMJ**

- occur in an **anterior direction** only as the **condyles slide down** unchecked along the slope of the articular eminence
- when this occurs, the jaw remains open and the person is unable to close it.
- This condition may result from:
 - a side blow to the chin
 - a fractured mandible
 - or spasm of the lateral pterygoid muscles causing an excessive contraction, as in taking an exceptionally large bite of food or from deep yawning



MUSCLES OF MASTICATION

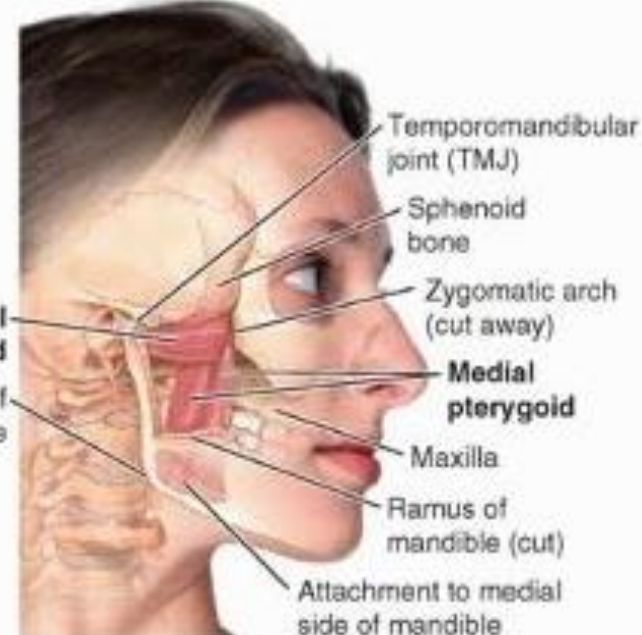
- TMJ movements are produced chiefly by the muscles of mastication.
- These four muscles:
 - Temporal
 - Masseter
 - and medial and lateral pterygoid muscles
 - develop from the mesoderm of the embryonic first pharyngeal arch
 - they are all innervated by the nerve of that arch, the (motor root of the) mandibular nerve (CN V₃)



(A) Masseter and temporalis

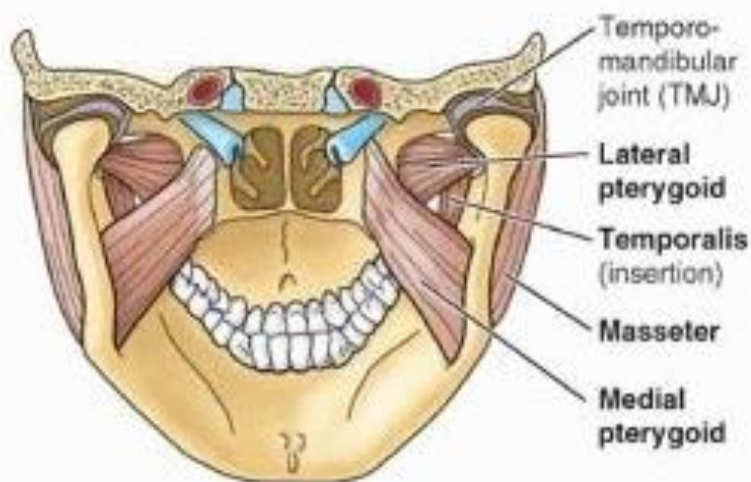


(B) Temporalis

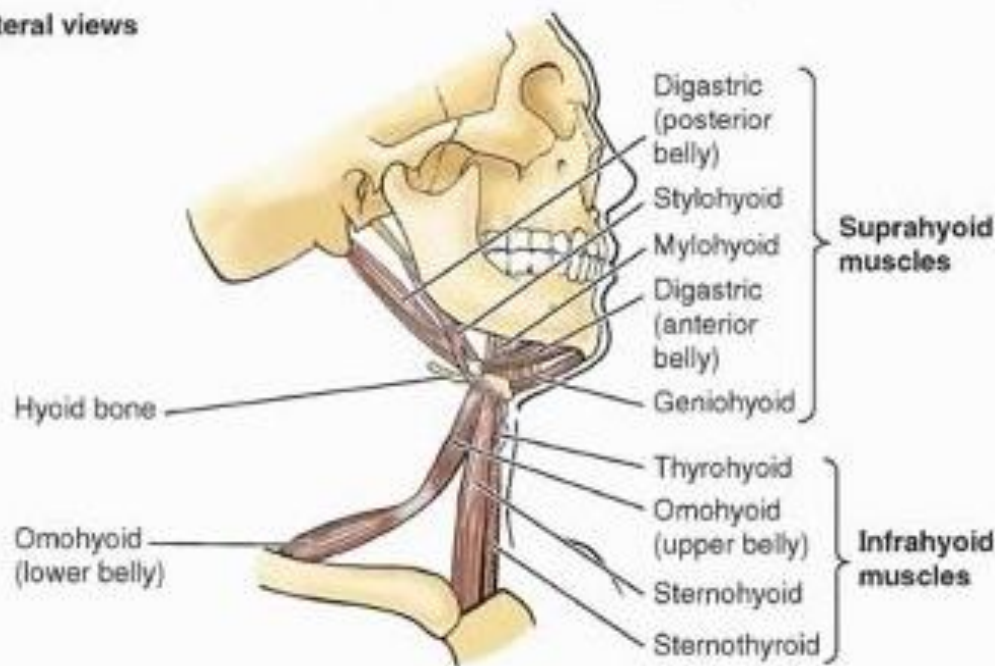


(C) Lateral and medial pterygoid

Lateral views



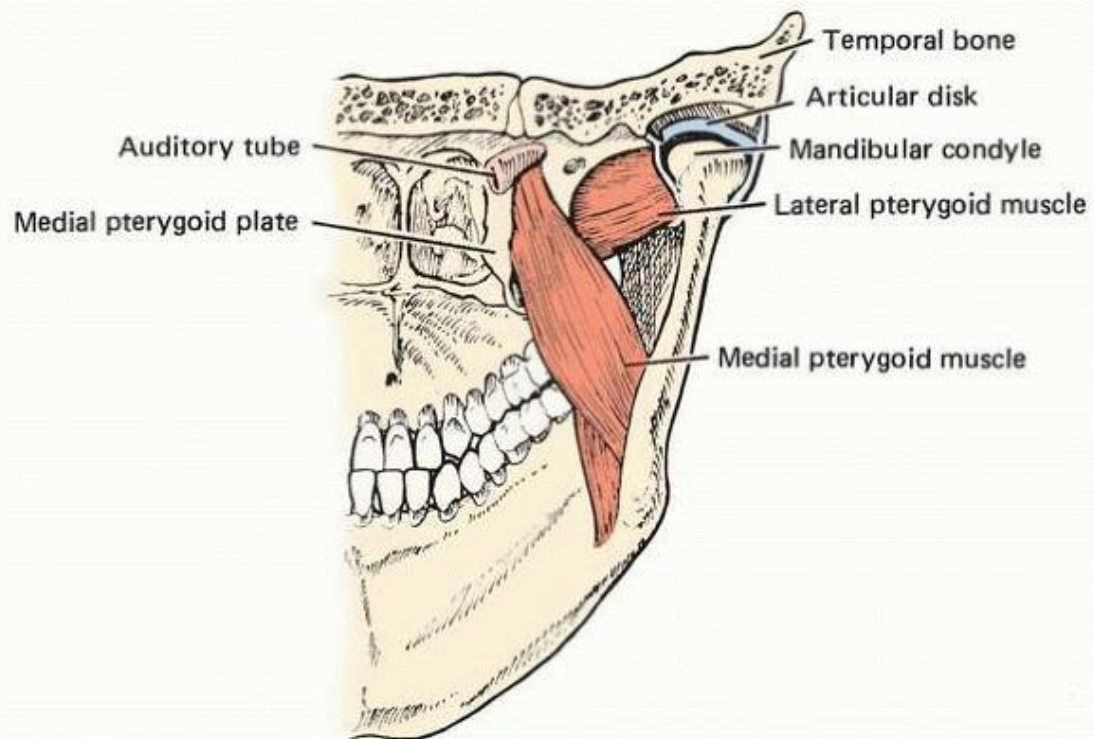
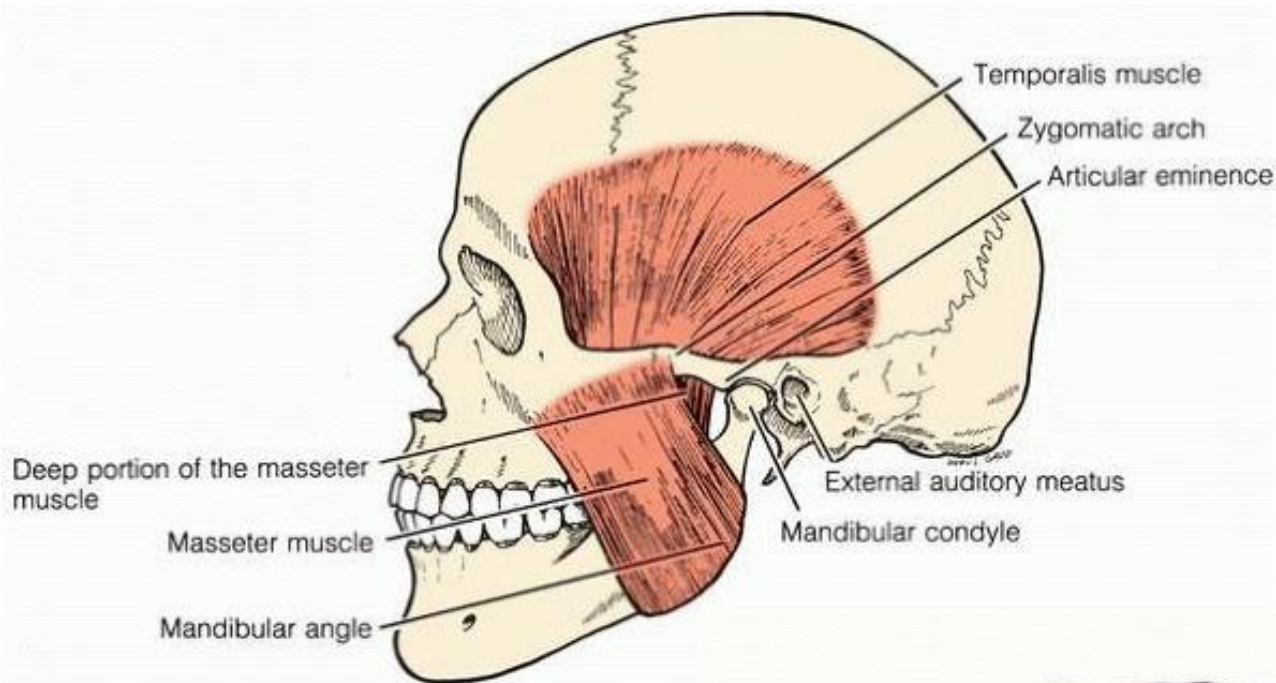
(D) Posterior view of viscerocranium

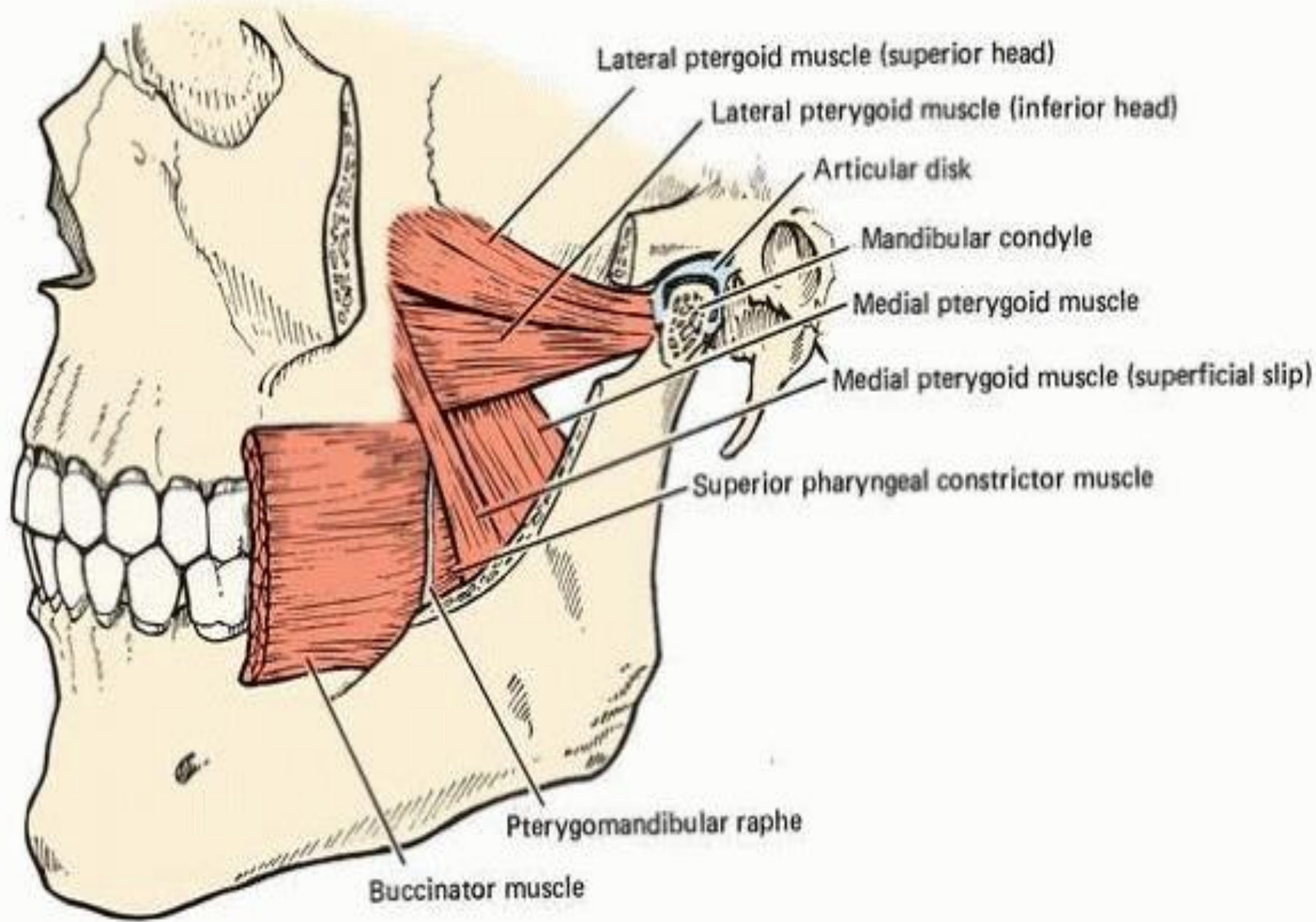


(E) Anterolateral view with head rotated slightly to left

Muscles of Mastication

Muscle	Origin	Insertion
Masseter	Superficial: tendinous aponeurosis from zygomatic process of maxilla and anterior two-thirds of inferior border of zygomatic arch Deep: medial aspect and inferior border of posterior one-third of zygomatic arch	Lateral aspect of ramus and angle of mandible as far anteriorly as the last molar tooth, as far superiorly as base of coronoid process
Temporalis	Inferior temporal line and bones of the temporal fossa	As a tendon on coronoid process and anterior border of ramus of mandible as far inferiorly and anteriorly as third molar
Medial (internal) pterygoid	Pterygoid fossa and medial surface of lateral pterygoid plate; one slip from lateral portion of pyramidal process of palatine bone and adjacent maxillary tuberosity	Medial surface of mandibular ramus as far superiorly as the sphenomandibular ligament, inferiorly to mylohyoid groove
Lateral (external) pterygoid	Superior head: greater wing of sphenoid and infratemporal crest Inferior head: lateral surface of lateral pterygoid plate	Superior head: articular capsule of temporomandibular joint—the disk and the superior portion of mandibular neck Inferior head: anterior surface of mandibular neck

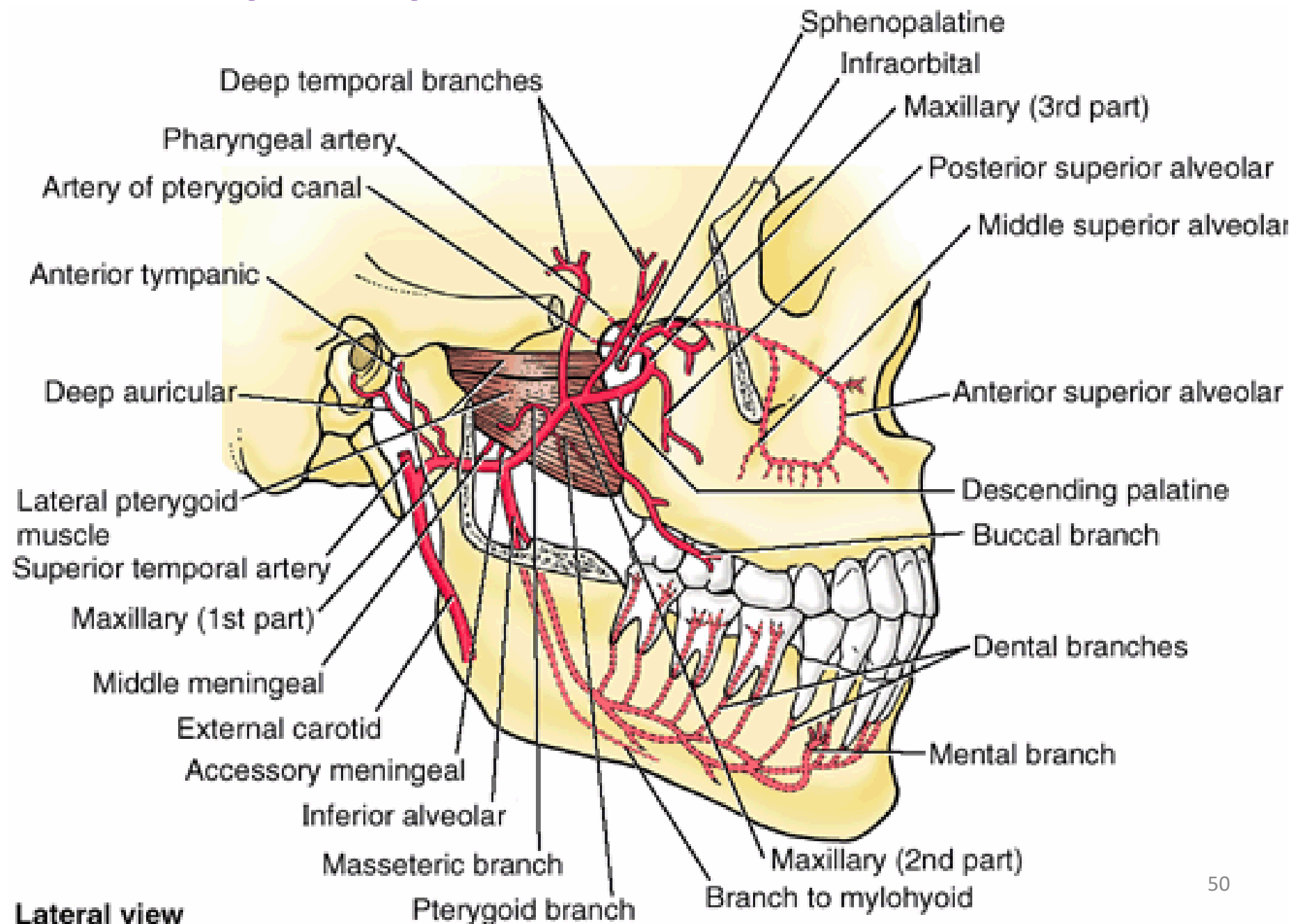




NEUROVASCULATURE OF INFRATEMPORAL FOSSA

- The maxillary artery:
 - is the larger of the two terminal branches of the external carotid artery
 - arises posterior to the neck of the mandible and is divided into three parts based on its relation to the lateral pterygoid muscle

• **Maxillary artery:**



First (mandibular) part

- Course:
 - Proximal (posterior) to lateral pterygoid muscle; runs horizontally, deep (medial) to neck of condylar process of mandible and lateral to stylomandibular ligament
- Branches:
 - **Deep auricular artery:**
 - Supplies external acoustic meatus, external tympanic membrane, and temporomandibular joint
 - **Anterior tympanic artery:**
 - Supplies internal aspect of tympanic membrane

- **Middle meningeal artery:**
 - Enters cranial cavity via **foramen spinosum**
 - supply periosteum, bone, red bone marrow, dura mater of lateral wall and calvaria of neurocranium, trigeminal ganglion, facial nerve and geniculate ganglion, tympanic cavity, and tensor tympani muscle
- **Accessory meningeal artery:**
 - Enters cranial cavity via **foramen ovale**
 - its distribution is mainly extracranial to muscles of infratemporal fossa, sphenoid bone, mandibular nerve, and otic ganglion

Inferior alveolar artery:

- Descends to enter mandibular canal of mandible via **mandibular foramen**
- supplies mandible, mandibular teeth, chin, mylohyoid

Second (pterygoid) part

- Course:
 - Adjacent (superficial or deep) to lateral pterygoid muscle
 - ascends obliquely anterosuperiorly, medial to temporal muscle

Branches:

- **Masseteric artery**
 - Traverses mandibular notch, supplying temporomandibular joint and masseter

- **Deep temporal arteries:**
 - Anterior and posterior arteries ascend between temporal muscle and bone of temporal fossa
 - supplying mainly muscle
- **Pterygoid branches:**
 - Irregular in number and origin
 - supply pterygoid muscle
- **Buccal artery:**
 - Runs anteroinferiorly with buccal nerve
 - supply buccal fat-pad, buccinator, and buccal oral mucosa

Third (pterygopalatine) part

- Distal (anteromedial) to lateral pterygoid muscle
- passes between heads of lateral pterygoid and through pterygomaxillary fissure into pterygopalatine fossa

Branches:

- **Posterior superior alveolar artery**

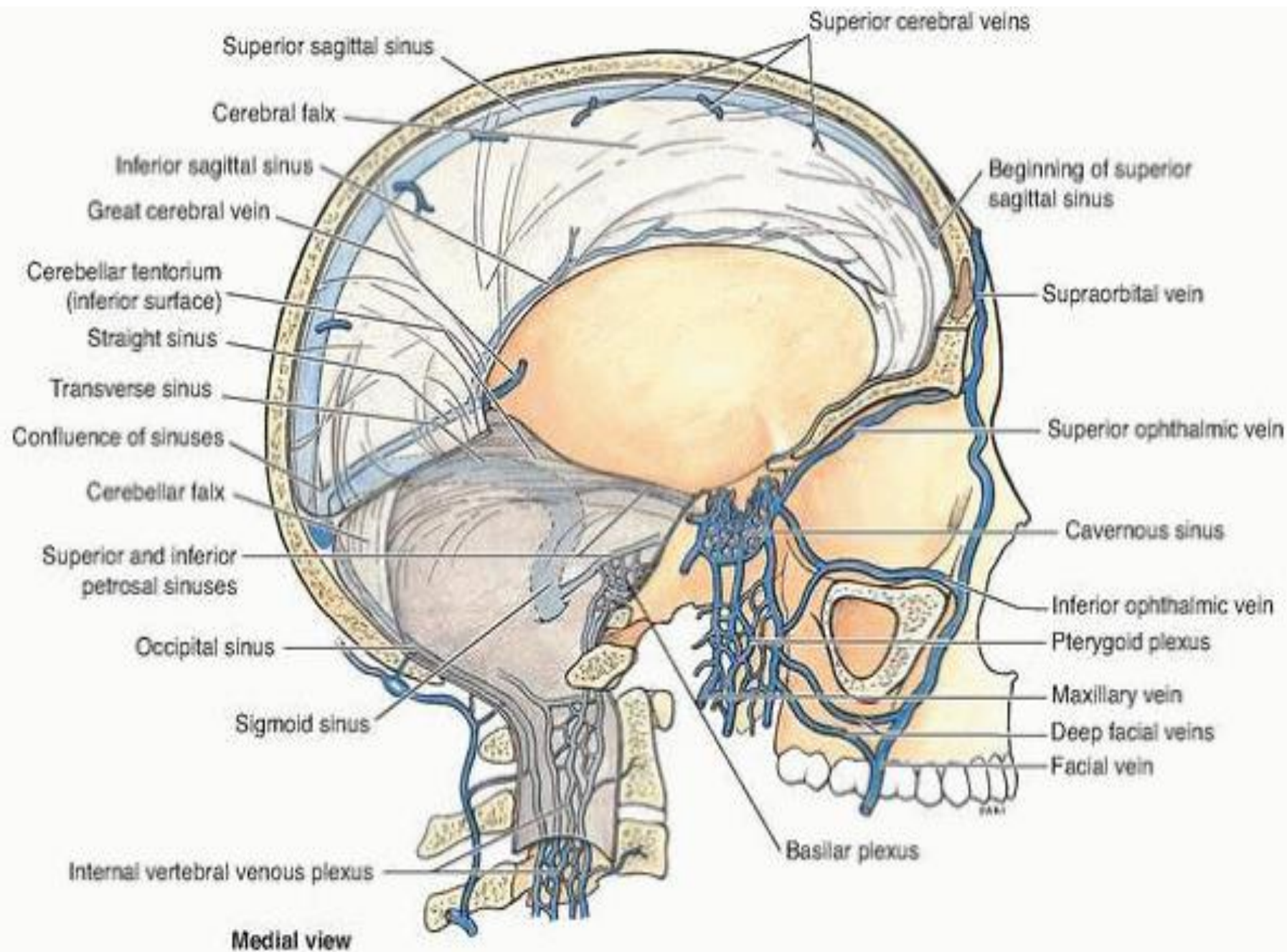
- Descends on maxilla's infratemporal surface with branches **traversing alveolar canals**
- supply **maxillary molar** and **premolar teeth**, adjacent gingiva, and mucous membrane of maxillary sinus

- **Infraorbital artery:**
 - Traverses inferior orbital fissure, infraorbital groove, canal, and foramen
 - supplies inferior oblique and rectus muscles, lacrimal sac, maxillary canines and incisors teeth, mucous membrane of the maxillary sinus, and skin of infraorbital region of face
- **Artery of pterygoid canal:**
 - Passes posteriorly through pterygoid canal
 - supplies mucosa of upper pharynx, pharyngotympanic tube, and tympanic cavity

- **Pharyngeal branch:**
 - Passes through palatovaginal canal
 - supply mucosa of nasal roof, nasopharynx, sphenoidal air sinus, and pharyngotympanic tube
- **Descending palatine artery:**
 - Descends through **palatine canal**, dividing into greater and lesser palatine arteries to mucosa and glands of hard and soft palate
- **Sphenopalatine artery:**
 - Terminal branch of maxillary artery
 - traverses sphenopalatine foramen to supply walls and septum of nasal cavity
 - frontal, ethmoidal, sphenoid, and maxillary sinuses; and anteriormost palate

- N.B. The parotid and temporal regions and the infratemporal fossa collectively include the temporomandibular joint and the muscles of mastication that produce its movements.

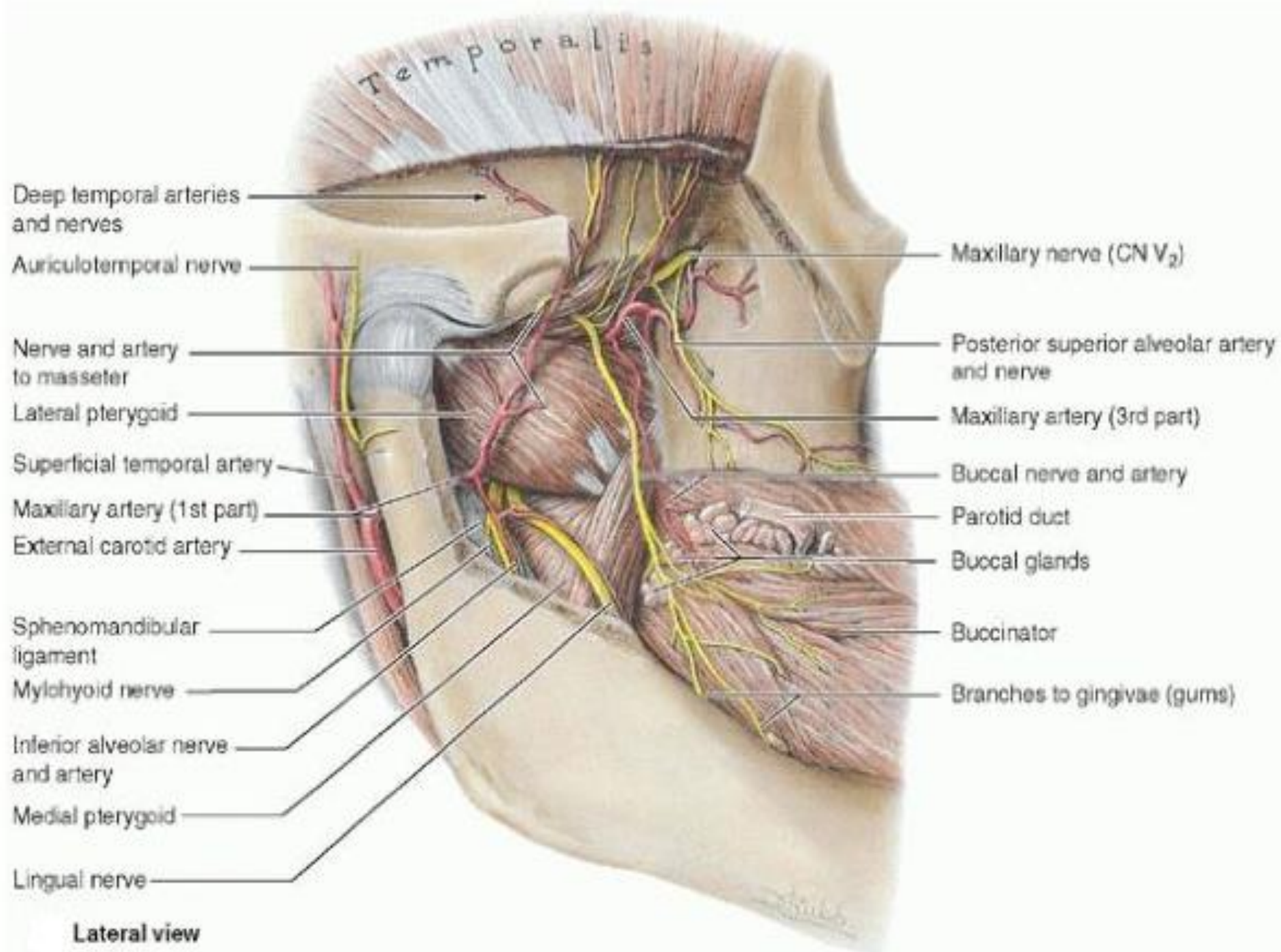
- **The pterygoid venous plexus:**
 - is located partly between the temporalis and the pterygoid muscles
 - It is the venous equivalent of most of the maxillary artery - that is, most of the veins that accompany the branches of the maxillary artery drain into this plexus.
 - The plexus anastomoses anteriorly with the facial vein via the deep facial vein and superiorly with the cavernous sinus via emissary veins.

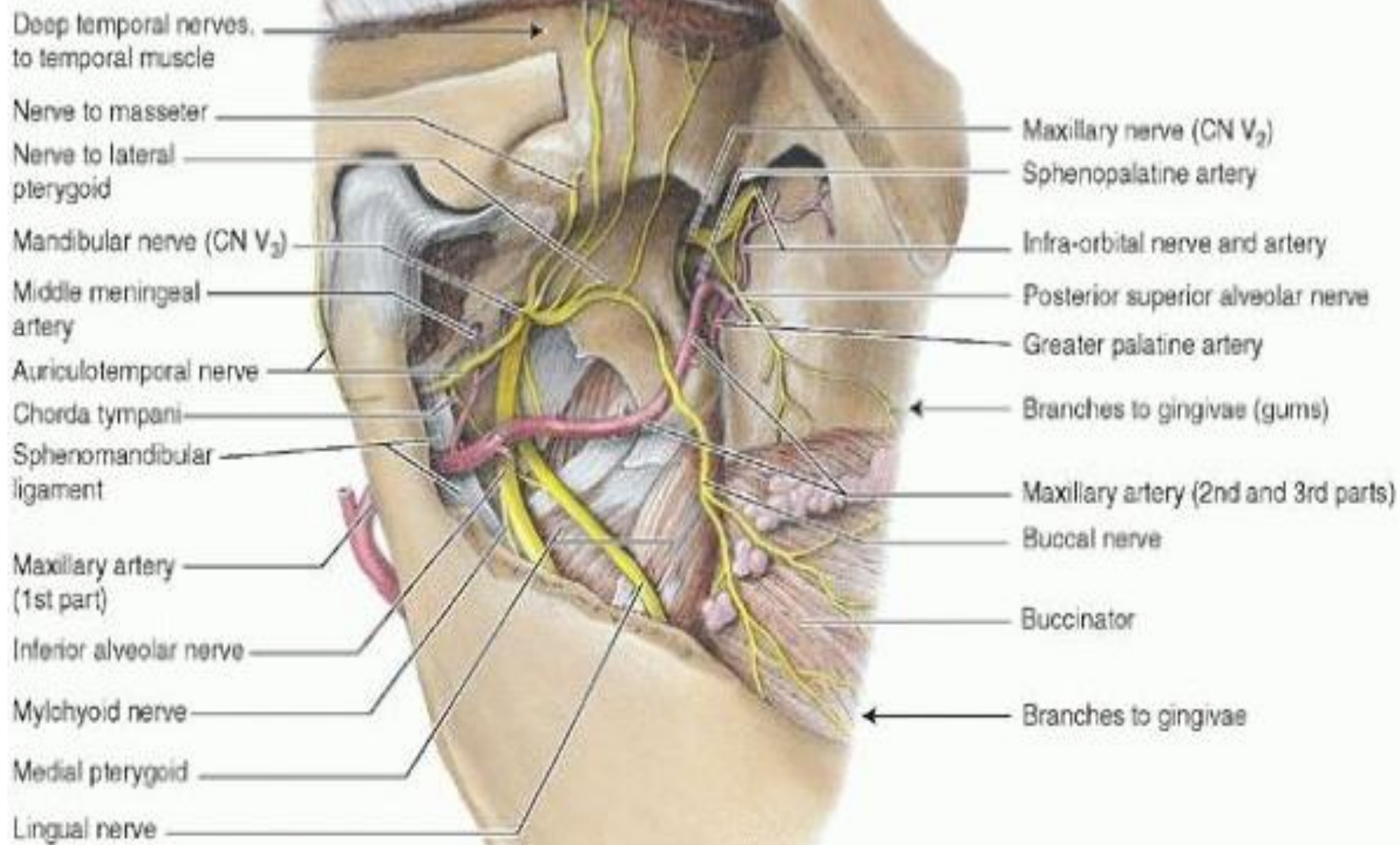


- The mandibular nerve:
 - arises from the trigeminal ganglion in the middle cranial fossa
 - It immediately receives the motor root of the trigeminal nerve and descends through the foramen ovale into the infratemporal fossa
 - Give rise to: the auriculotemporal, inferior alveolar, lingual, and buccal nerves.
 - Branches of CN V3 also supply the four muscles of mastication but not the buccinator, which is supplied by the facial nerve.

- **The auriculotemporal nerve:**
 - encircles the middle meningeal artery and divides into numerous branches, the largest of which passes posteriorly, medial to the neck of the mandible, and supplies sensory fibers to the auricle and temporal region
 - The auriculotemporal nerve also sends articular (sensory) fibers to the TMJ
 - It conveys postsynaptic parasympathetic secretomotor fibers from the otic ganglion to the parotid gland.
- **The inferior alveolar nerve:**
 - enters the mandibular foramen and passes through the mandibular canal, forming the inferior dental plexus, which sends branches to all mandibular teeth on its side.
 - Another branch of the plexus, the mental nerve, passes through the **mental foramen** and supplies the skin and mucous membrane of the lower lip, the skin of the chin, and the vestibular gingiva of the mandibular incisor teeth

- **The lingual nerve:**
 - lies anterior to the inferior alveolar nerve
 - It is sensory to the anterior two thirds of the tongue, the floor of the mouth, and the lingual gingivae.
 - It enters the mouth between the **medial pterygoid muscle** and the ramus of the mandible and passes anteriorly under cover of the oral mucosa, just inferior **to the 3rd molar tooth**.
- **The chorda tympani nerve:**
 - a branch of CN VII carrying taste fibers from the anterior two thirds of the tongue
 - joins the lingual nerve in the infratemporal fossa
 - carries **secretomotor fibers** for the submandibular and sublingual salivary glands

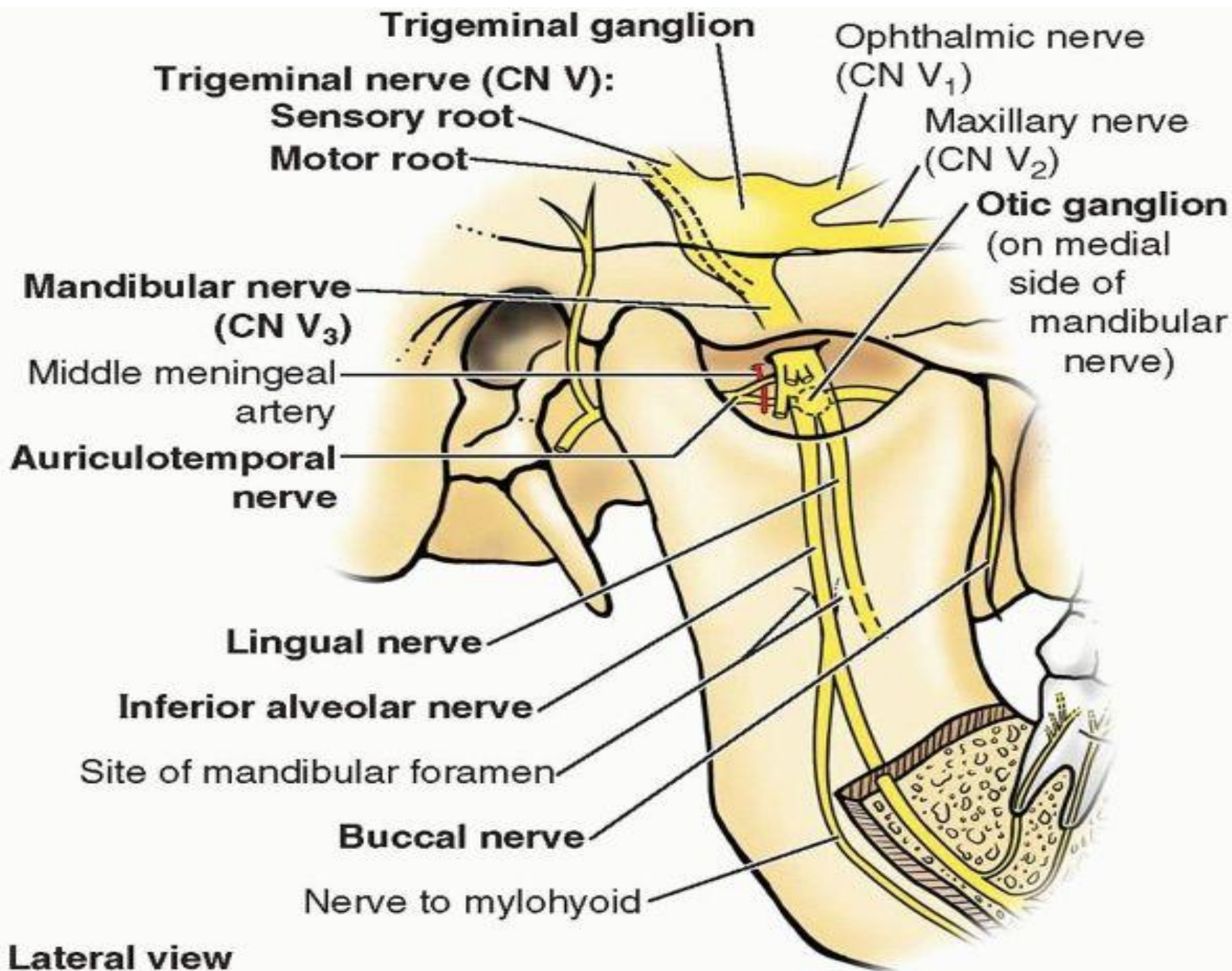




(B) Lateral view

- **The otic ganglion (parasympathetic):**

- is located in the infratemporal fossa,
 - just inferior to the foramen ovale
 - medial to CN V3
 - posterior to the medial pterygoid muscle
- Presynaptic parasympathetic fibers, derived mainly from the glossopharyngeal nerve, synapse in the otic ganglion
- Postsynaptic parasympathetic fibers pass from the otic ganglion to this gland through the auriculotemporal nerve



Pterygopalatine Fossa, Nasal Cavity, and Paranasal Sinuses

Pterygopalatine Fossa

- is a small pyramidal space inferior to the apex of the orbit
- lies between:
 - The pterygoid process of the sphenoid posteriorly
 - The posterior aspect of the maxilla anteriorly.
 - The medial wall is formed of perpendicular plate of the palatine bone
 - The roof:
 - The incomplete and is formed by the greater wing of the sphenoid.
 - The floor:
 - formed by the pyramidal process of the palatine bone.
 - Its superior larger end opens into the inferior orbital fissure
 - its inferior end is closed except for the palatine foramina.

The pterygopalatine fossa communicates:

- Laterally with the infratemporal fossa through the pterygomaxillary fissure.
- Medially with the nasal cavity through the sphenopalatine foramen.
- Anterosuperiorly with the orbit through the inferior orbital fissure.
- Posterosuperiorly with the middle cranial fossa through the foramen rotundum and pterygoid canal

The contents

- Terminal (pterygopalatine or third) part of the maxillary artery and the initial parts of its branches.
- Maxillary nerve, with which are associated the:
 - Nerve of the pterygoid canal.
 - Pterygopalatine ganglion.

The Nose

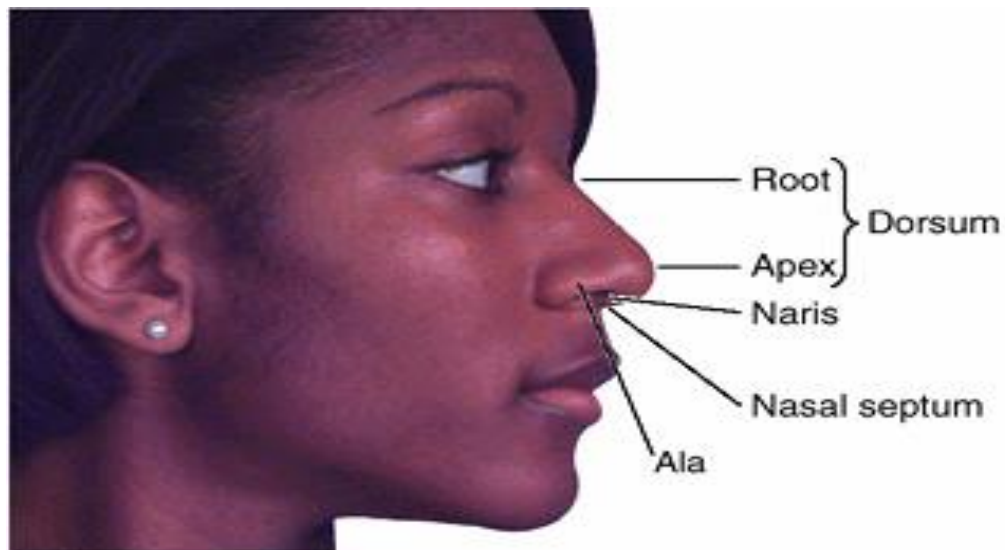
- the part of the respiratory tract superior to the hard palate
- contains the peripheral organ of smell.
- **It includes:**
 - the external nose and nasal cavity
 - The functions of the nose are olfaction (smelling), respiration (breathing), filtration of dust, humidification of inspired air, and reception and elimination of secretions from the paranasal sinuses and nasolacrimal ducts

External Nose

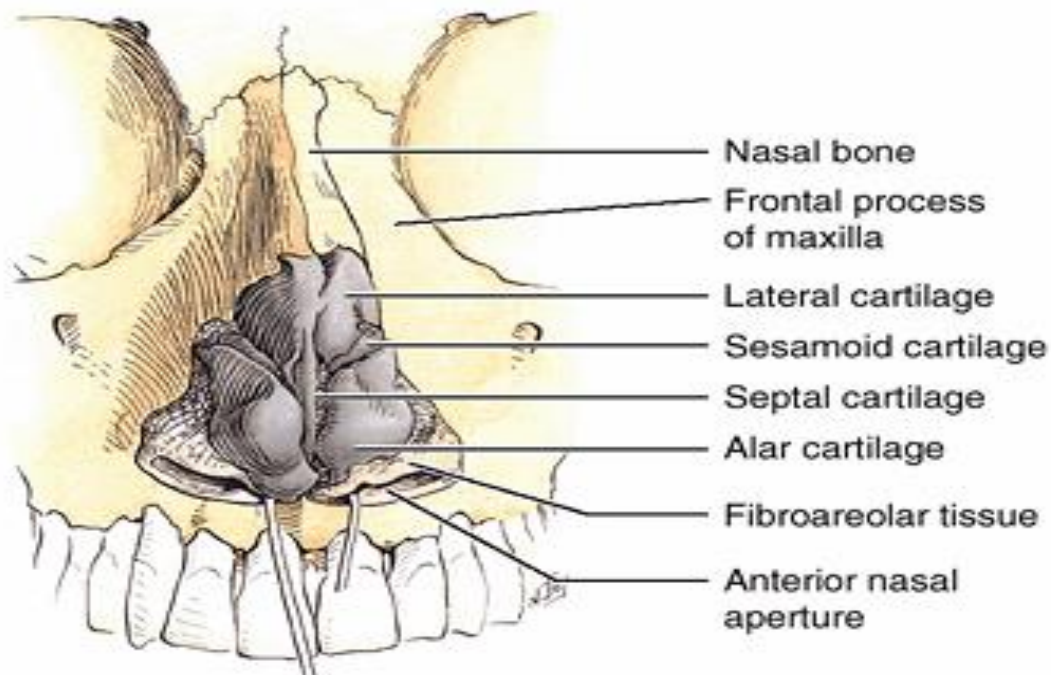
- has two elliptical orifices called the nostrils
- The lateral margin, the ala nasi, is rounded and mobile.
- The framework of the external nose is made up of **above by**
 - the nasal bones
 - the frontal processes of the maxillae
 - the nasal part of the frontal bone.
- **Below**, the framework is formed of plates of hyaline cartilage

Skeleton of the External Nose

- composed of bone and hyaline cartilage.
- The bony part of the nose consists of:
 - the nasal bones
 - frontal processes of the maxillae
 - the nasal part of the frontal bone
 - its nasal spine, and the bony parts of the nasal septum.
- The cartilaginous part of the nose consists of five main cartilages:
 - two lateral cartilages
 - two alar cartilages
 - one septal cartilage.
 - The U-shaped alar cartilages are free and movable
 - they dilate or constrict the nares when the muscles acting on the nose contract



(A) Lateral view



(B) Anterior view

Nasal Cavity

- extends from the **nostrils** in front to the posterior nasal apertures or **choanae** behind
 - The nasal vestibule is the area of the nasal cavity lying just inside the nostril
 - divided into right and left halves by the nasal septum
 - The septum is made up of the septal cartilage, the vertical plate of the **ethmoid**, and **the vomer**

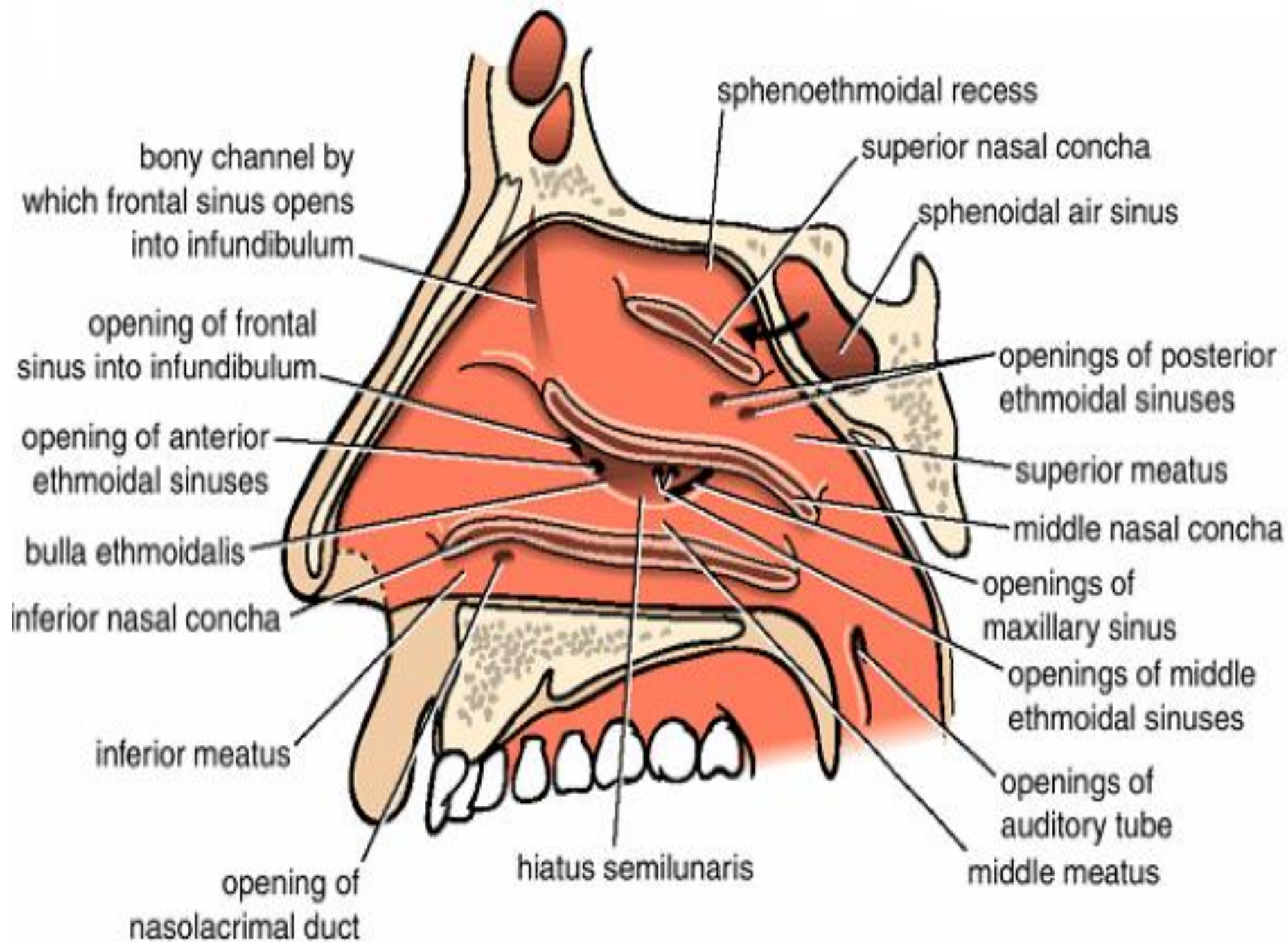
Walls of the Nasal Cavity

- Each half of the nasal cavity has:
 - a floor
 - a roof
 - a lateral wall, and a medial or septal wall.
- **Floor**
 - The palatine process of the maxilla and the horizontal plate of the palatine bone

- **Roof**
 - narrow and is formed **anteriorly by** :
 - the nasal and frontal bones
 - **in the middle by:**
 - the cribriform plate of the ethmoid
 - located beneath the anterior cranial fossa
 - **and posteriorly by:**
 - the downward sloping body of the sphenoid

Lateral Wall

- The lateral wall has three projections of bone called:
 - The superior
 - Middle
 - and inferior nasal conchae
 - The space below each concha is called a meatus



Sphenoethmoidal Recess

- is a small area above the superior concha
- It receives the opening of the sphenoid air sinus

Superior Meatus

- lies below the superior concha
- It receives the openings of the posterior ethmoid sinuses.

Middle Meatus

- lies below the middle concha
- It has a rounded swelling called the **bullae ethmoidalis** that is formed by the **middle ethmoidal air sinuses**, which open on its upper border.
- A curved opening, **the hiatus semilunaris**, lies just below the bulla
- The anterior end of the **hiatus leads** into a funnel-shaped channel called **the infundibulum**, which is continuous with the frontal sinus.
- The maxillary sinus opens into the middle meatus through the **hiatus semilunaris**

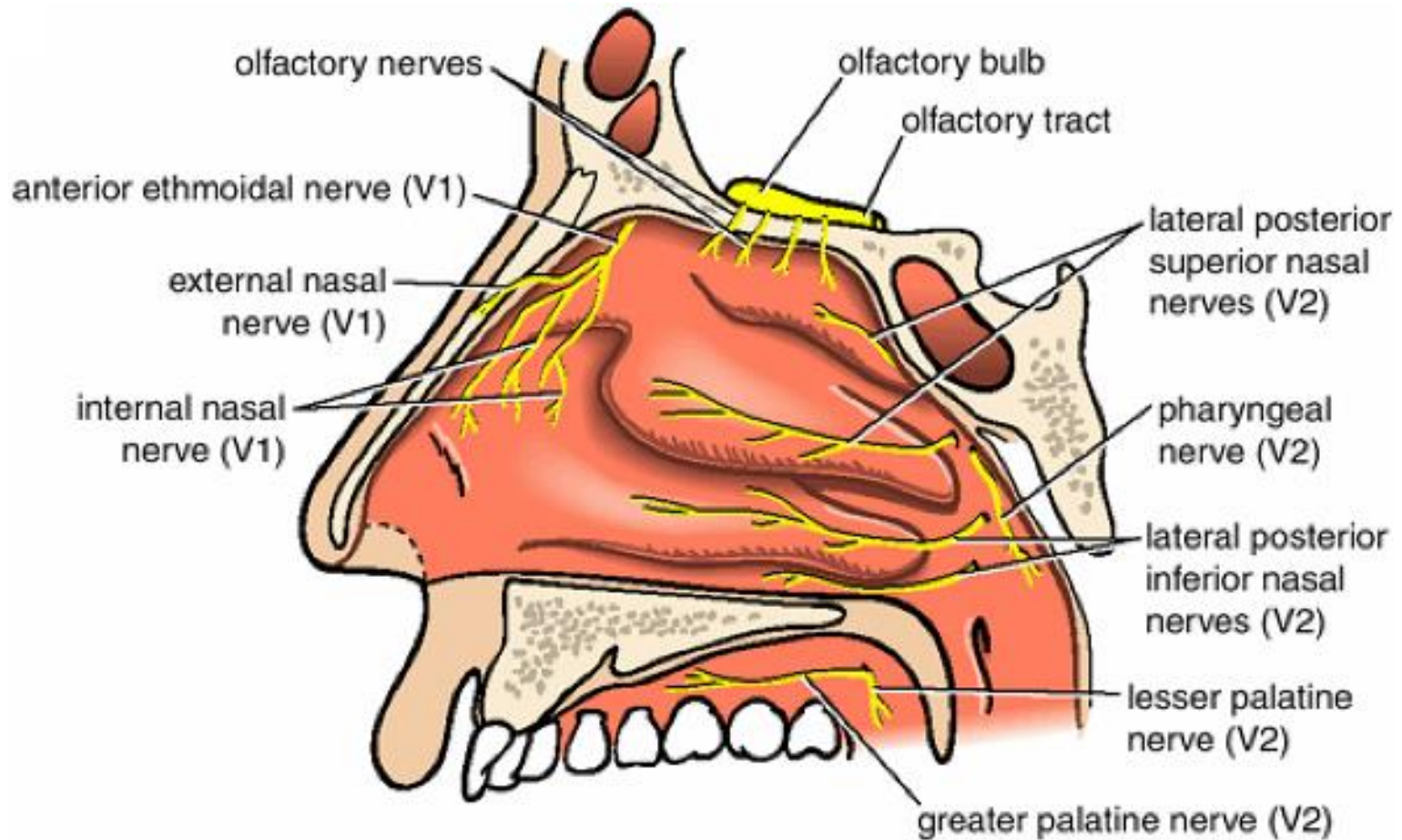
Inferior Meatus

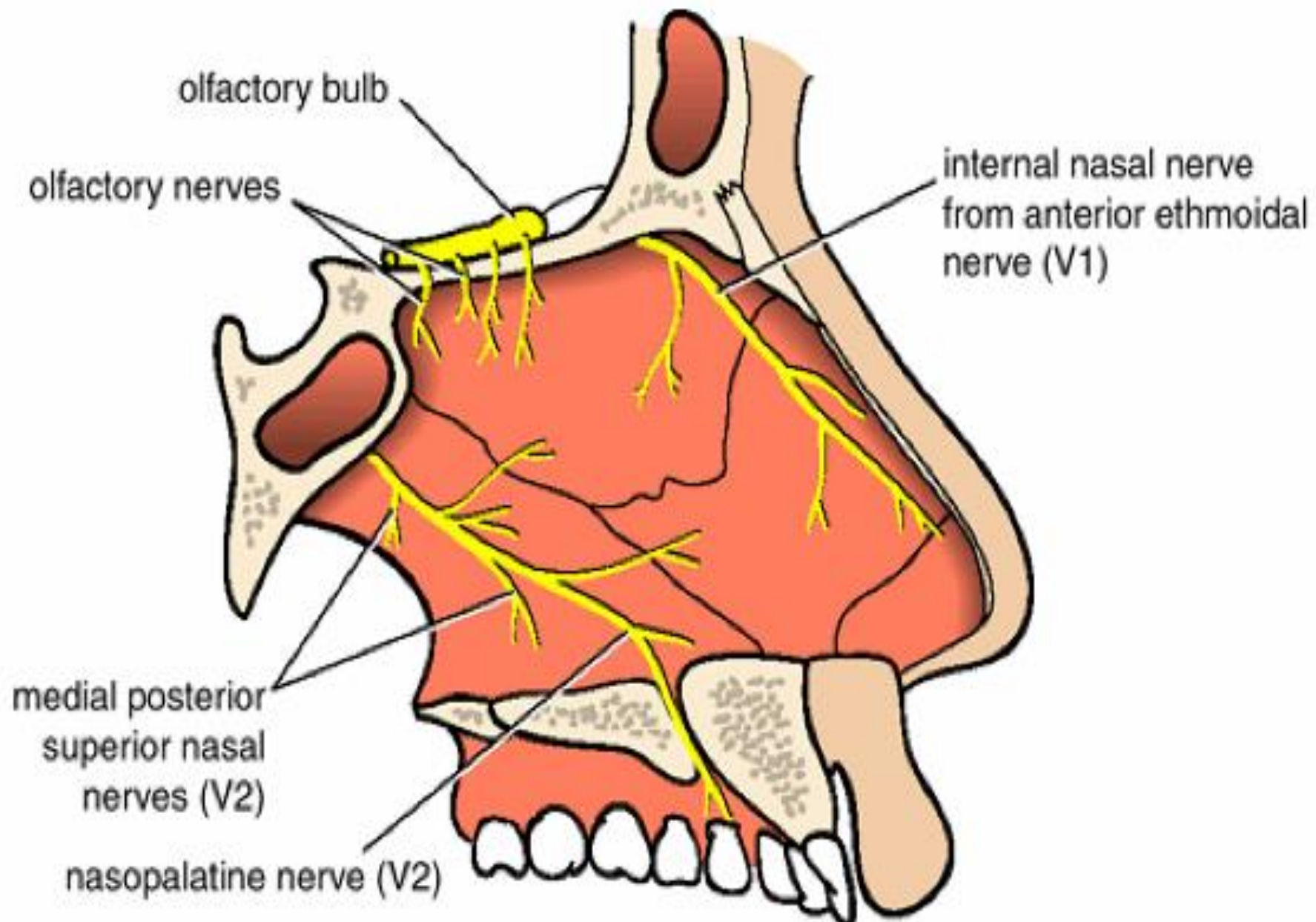
- lies below the inferior concha
- receives the opening of the lower end of the nasolacrimal duct, which is guarded by a fold of **mucous membrane**

Medial Wall

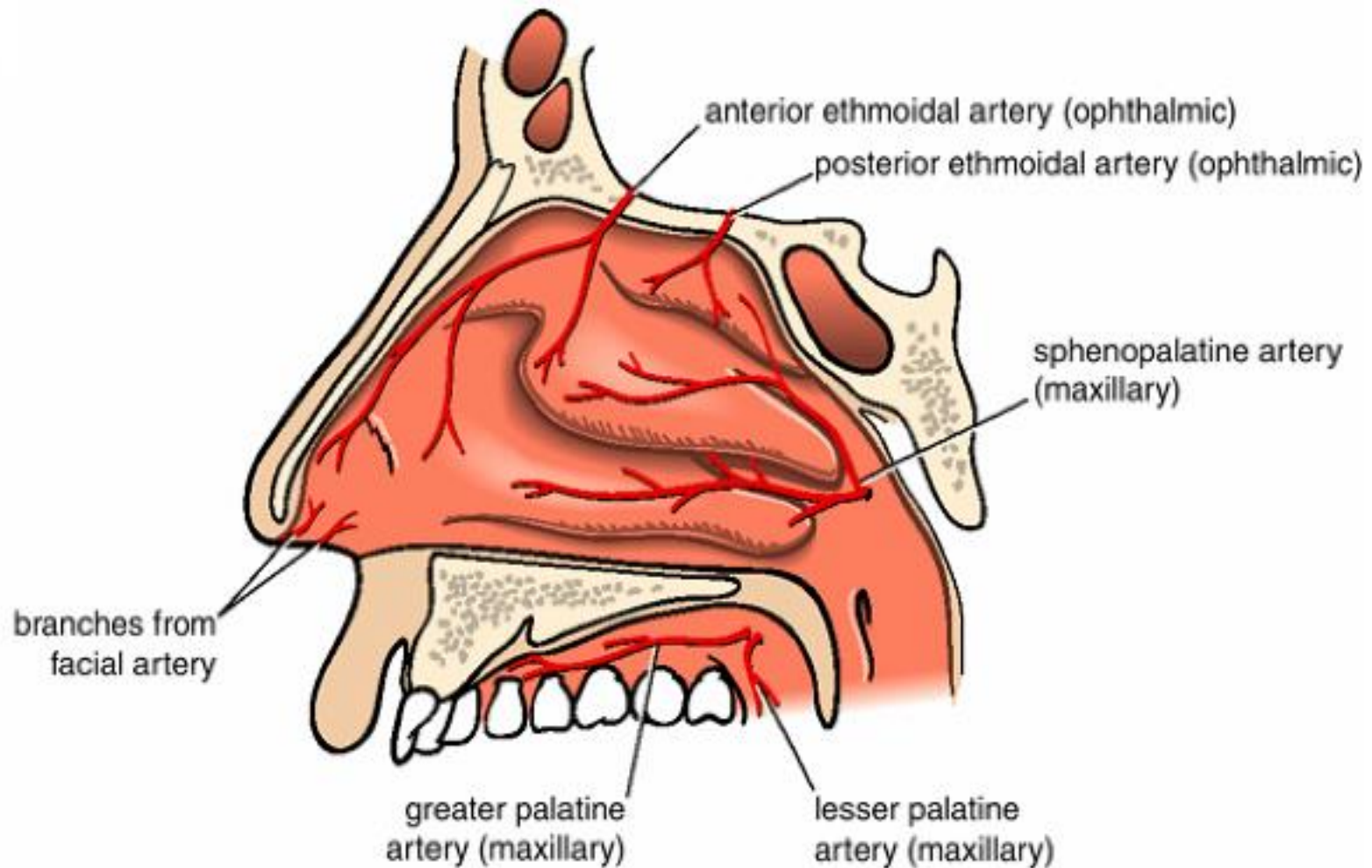
- formed by the nasal septum.
- The upper part is formed by:
 - the vertical plate of the ethmoid and the vomer
- The anterior part is formed by:
 - the septal cartilage.
 - The septum rarely lies in the midline, thus increasing the size of one half of the nasal cavity and decreasing the size of the other

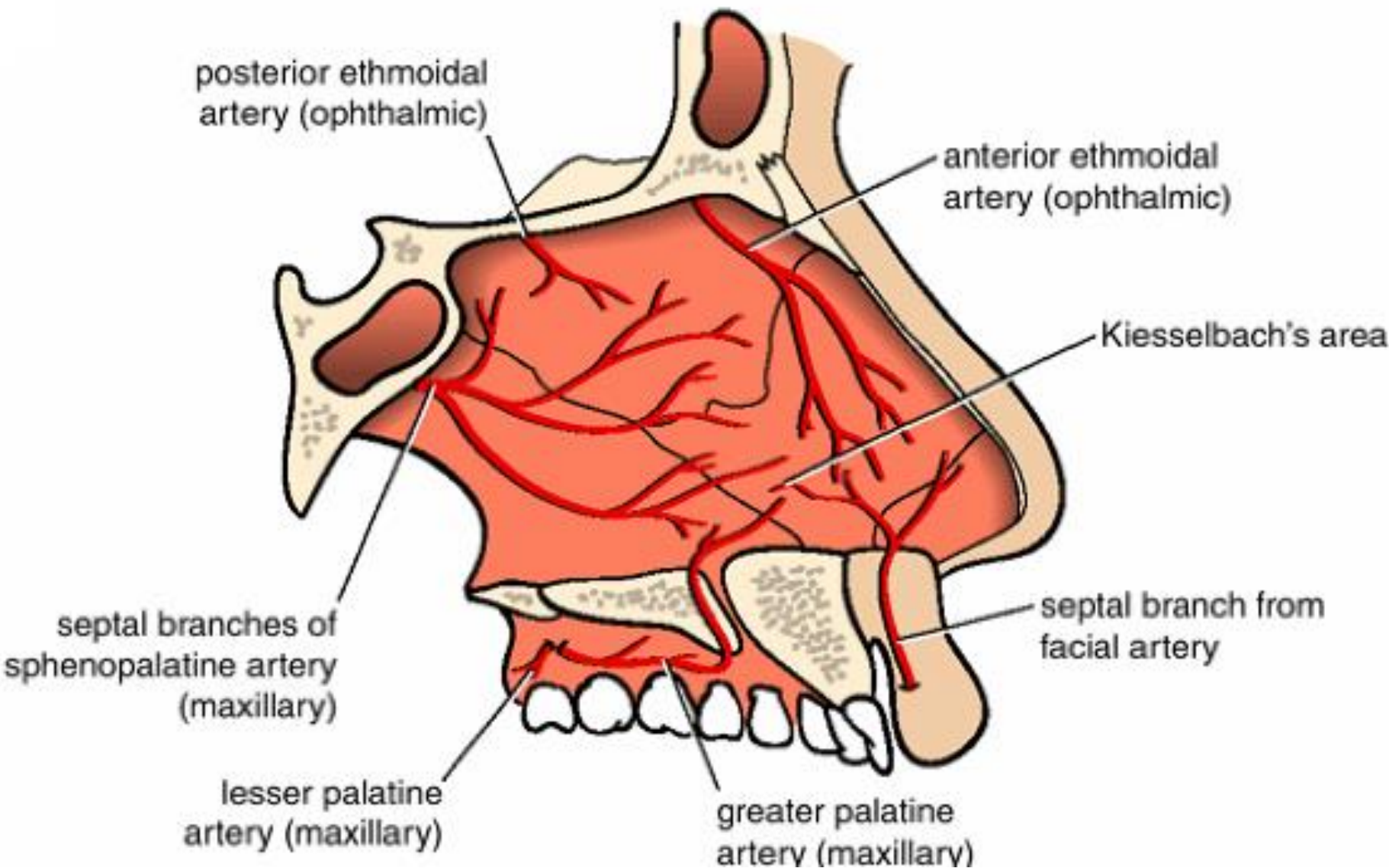
Nerve supply

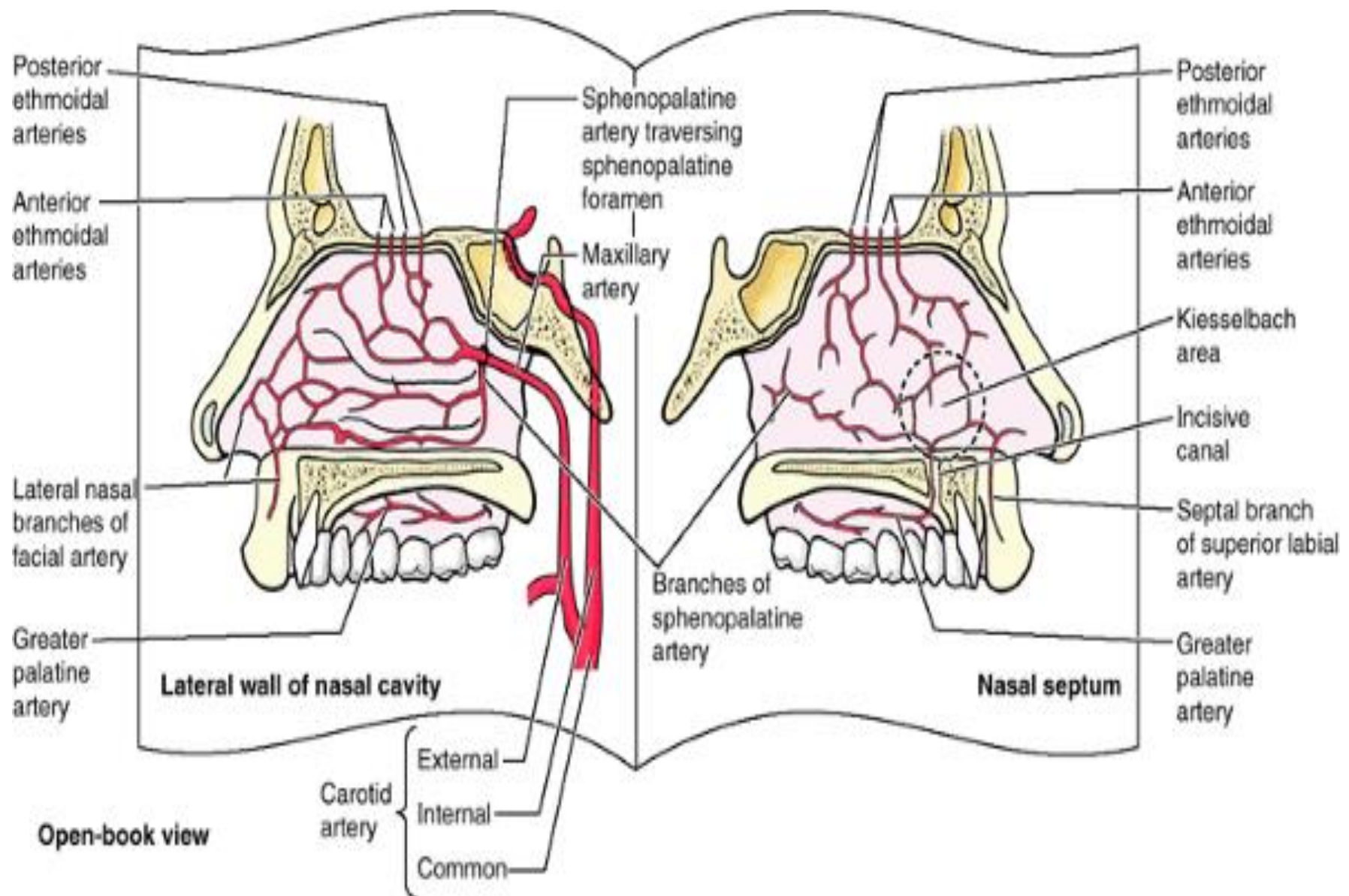




Blood supply







- A rich submucosal venous plexus deep to the nasal mucosa drains into:
 - the sphenopalatine
 - facial, and ophthalmic veins.
 - This **venous plexus** is an important part of the body's thermoregulatory system, exchanging heat and warming air before it enters the lungs.
 - Venous blood from the external nose drains mostly into **the facial vein** via **the angular** and **lateral nasal veins**.
 - However, recall that it lies within **the danger area** of the face because of communications with the cavernous (**dural venous**) sinus

Rhinitis

- The nasal mucosa becomes swollen and inflamed (rhinitis) during severe upper respiratory infections and allergic reactions (e.g., hayfever). Swelling of the mucosa occurs readily because of its vascularity.
- Infections of the nasal cavities may spread to the:
 - Anterior cranial fossa through the cribriform plate.
 - Nasopharynx and retropharyngeal soft tissues.
 - Middle ear through the pharyngotympanic tube (auditory tube), which connects the tympanic cavity and nasopharynx.
 - Paranasal sinuses.
 - Lacrimal apparatus and conjunctiva

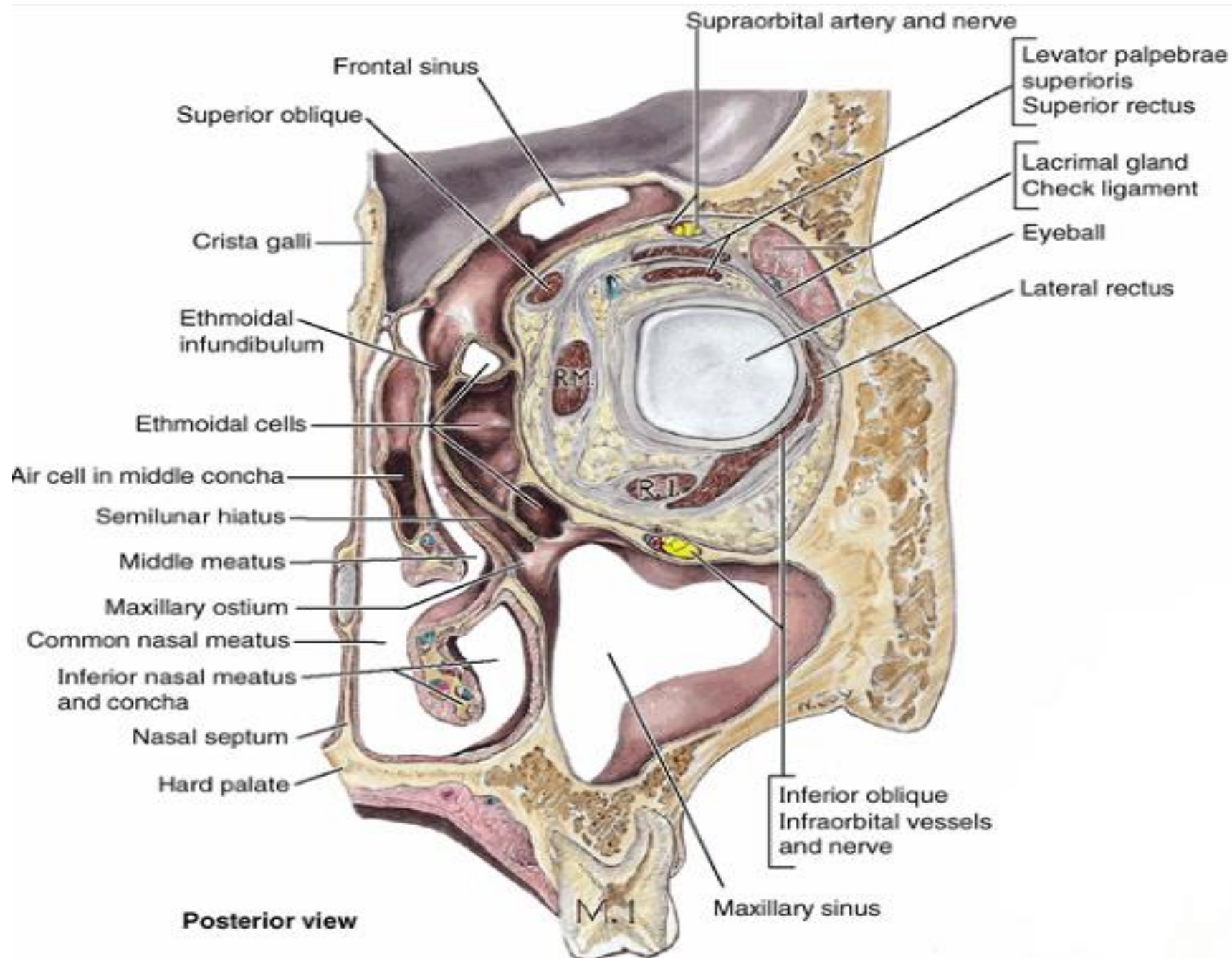
Paranasal sinus (four)

Frontal Sinuses

- are between the outer and the inner tables of the frontal bone, posterior to the superciliary arches and the root of the nose
- usually detectable in children by 7 years of age
- Each sinus drains through a frontonasal duct into the **ethmoidal infundibulum**-semilunar hiatus of the middle nasal meatus.
- innervated by branches of the **supraorbital nerves** (CN V₁).

Ethmoidal Cells

- small invaginations of the mucous membrane of the middle and superior nasal meatus into the ethmoid bone between the nasal cavity and the orbit
- The ethmoidal cells usually are not visible in plain radiographs before 2 years of age but are recognizable in CT scans.
- The **anterior ethmoidal cells** drain directly or indirectly into the middle nasal meatus through the **ethmoidal infundibulum**.
- **The middle ethmoidal cells** open directly into the middle meatus and are sometimes **called bullar cells** because they form the **ethmoidal bulla**, a swelling on the superior border of the semilunar hiatus
- **The posterior ethmoidal cells open** directly into the superior meatus.
- The ethmoidal cells are supplied by the **anterior** and **posterior ethmoidal** branches of **the nasociliary nerves** (CN V₁)



Infection of the Ethmoidal Cells

- If nasal drainage is blocked, infections of the ethmoidal cells may break through the fragile medial wall of the orbit
- Severe infections from this source may cause blindness because some posterior ethmoidal cells lie close to the optic canal, which gives passage to the optic nerve and ophthalmic artery.
- Spread of infection from these cells could also affect the dural nerve sheath of the optic nerve, causing optic neuritis

Sphenoidal Sinuses

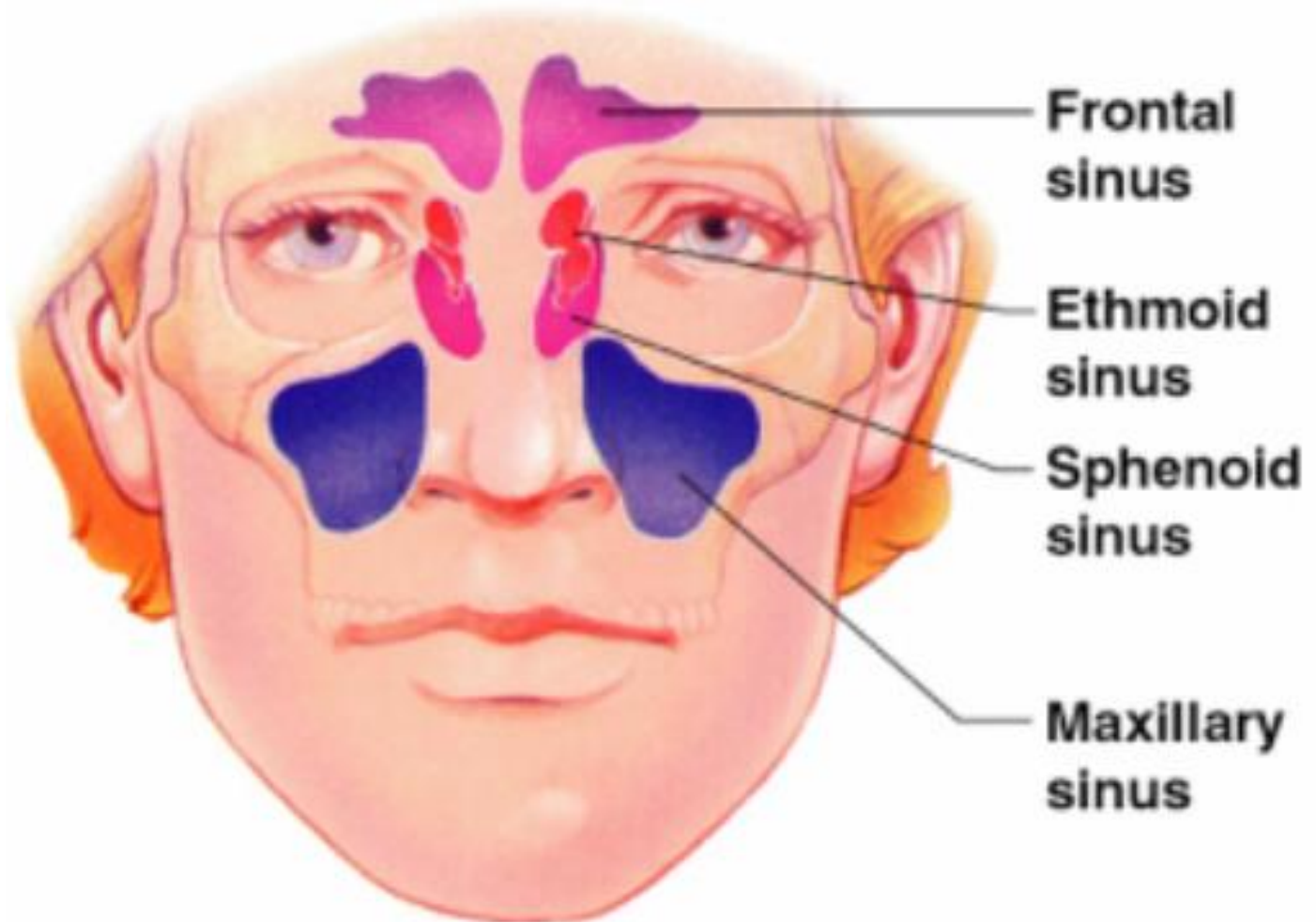
- located in the body of the sphenoid and may extend into the wings of this bone
- are unevenly divided and separated by a bony septum.
- Because of this extensive pneumatization (formation of air cells or sinuses), the body of the sphenoid is fragile.
- Only thin plates of bone separate the sinuses from several important structures:
 - the optic nerves and optic chiasm
 - the pituitary gland
 - the internal carotid arteries
 - and the cavernous sinuses.

- The sphenoidal sinuses are derived **from a posterior ethmoidal cell** that begins to invade the sphenoid at **approximately 2 years of age**.
- In some people, several posterior ethmoidal cells invade the sphenoid, giving rise to multiple sphenoidal sinuses that open separately into the **sphenoethmoidal recess**.
- The posterior ethmoidal arteries and posterior ethmoidal nerve supply the sphenoidal sinuses

Maxillary Sinuses

- The maxillary sinuses are the **largest of the paranasal sinuses**.
- They occupy the bodies of the maxillae and communicate with the middle nasal meatus
 - **The apex** : extends toward and often into the zygomatic bone.
 - **The base**: forms the inferior part of the lateral wall of the nasal cavity.
 - **The roof** is formed by the floor of the orbit.
 - **The floor** is formed by the alveolar part of the maxilla.
 - The roots of the maxillary teeth, particularly the first two molars, often produce conical elevations in the floor of the sinus.

- Each maxillary sinus drains by one or more openings
 - the maxillary ostium (ostia), into the middle nasal meatus of the nasal cavity by way of the **semilunar hiatus**

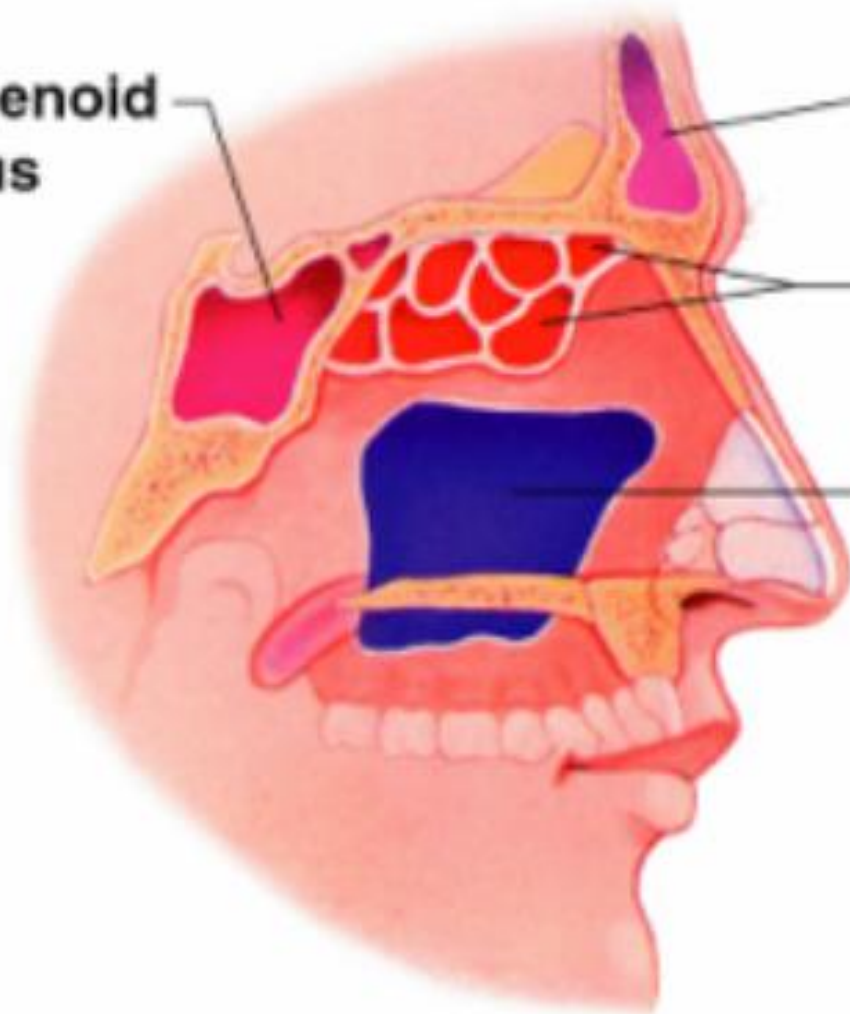


**Sphenoid
sinus**

**Frontal
sinus**

**Ethmoid
sinus**

**Maxillary
sinus**



- The arterial supply of the maxillary sinus is:
 - mainly from superior alveolar branches of the maxillary artery
 - branches of the descending and greater palatine arteries supply the floor of the sinus
- Innervation of the maxillary sinus is from:
 - the anterior superior alveolar nerves
 - middle superior alveolar nerves
 - posterior superior alveolar nerves, which are branches of the maxillary nerve

Infection of the Maxillary Sinuses

- the most commonly infected, **why?**
- Because the ostia of the right and left sinuses lie on the medial sides (i.e., are directed toward each other), when lying on one's side only the upper sinus (e.g., the right sinus if lying on the left side) drains.
- A cold or allergy involving both sinuses can result in nights of rolling from side-to-side in an attempt to keep the sinuses drained.
- A maxillary sinus can be cannulated and drained by passing a cannula from the nares through the maxillary ostium into the sinus.

Relationship of the Teeth to the Maxillary Sinus

- The close proximity of the three maxillary molar teeth to the floor of the maxillary sinus poses potentially serious problems
- During removal of a molar tooth, a fracture of a root may occur.
- If proper retrieval methods are not used, a piece of the root may be driven superiorly into the maxillary sinus.
- A communication may be created between the oral cavity and the maxillary sinus as a result, and an infection may occur
- Because the superior alveolar nerves (branches of the maxillary nerve) supply both the maxillary teeth and the mucous membrane of the maxillary sinuses, inflammation of the mucosa of the sinus is frequently accompanied by a sensation of toothache in the molar teeth

Thank You!